# ESSS

## Employee Self Service System

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**Project Of:**

**Tim Coker & John Varnadore**

**April 2002**

**Under the Supervision of:**

**Dr. Jack Briner**
**Doctor of Computer Science**
**Charleston Southern University**
**(843) 863-7529**

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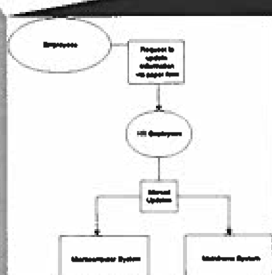
## History Leading to Proposal for Project

The current system is a combination of a mainframe system and a microcomputer. Employees send requests to Human Resources to update their information via paper based forms. The HR employees must enter the data on the mainframe and microcomputer systems. This process is expensive and is prone to error. The mainframe is used for administrative type duties (payroll, etc.) while the microcomputer is used solely to produce a phone book. This phone book is almost out of date once it is printed. Given the current level of growth experienced by the company, keeping data up to date has become a overwhelming problem.

## Existing System



The employees submit a request to update their information via a paper to the Human Resources dept. HR employee must then enter the data twice in the mainframe and microcomputer systems.

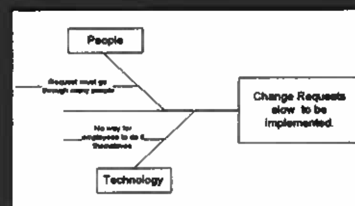
## Scope of the Project

The main business area that will be affected by this system will be the Human Resources, though all employees will ultimately use the system. It will completely replace the existing mainframe and microcomputer systems and will effectively offload the work of maintaining the employees' information to the employees themselves. It will provide an interface where employees can easily change the address, phone number and other personal information that the company has on file from any location and/or workstation.

## Problems

- Telephone book in hardcopy form, and it is difficult to update the information.
- Micro and mainframe both used   when possible to maybe just use one when employee information is being entered into the system.
- Personnel data maybe outdated or incorrect throughout the entire company systems.
- Different databases have duplicate employee information.
- Because of the information not being updated probably, the company mailings are incorrect and payroll checks are not delivered. Employee not able to locate other employee's information quickly and sufficiently.
- Users are unable to access reports in a sufficient time period.

## Problem Diagram




## Causes of Problems

- Personnel information not stored in the same source repository.
- Administrator keys in information after being submitted by the users on forms causing a delay in the information being stored.
- The Information entered is not in real-time, the information doesn't actually get out to the users until the new quarterly hardcopy telephone book is published.
- There is not a place that employee information is located that other employees can access their information.
- The IS personnel cannot keep up with the request of the employees in a time efficient manner, because of backup of request and other job priorities.

## Project Objectives

- Eliminate the Telephone books from being in hard copy form.
- Create a contact list that is online, so it would be easily accessible, to all users.
- Cut down on the time that the administrator has to maintain the system.
- Eliminate the duplication of information.
- Have employees be able to enter and update their personal information themselves.
- Make a single source repository to store the employee information.
- Have real-time changes to information inputted by the employees.
- Have online employee listing accessible to all employees.

## Constraints

- The system must be accessible from remote locations and desktops alike.
- Must be secure so unauthorized data cannot be added to the system.
- The time frame is three months.
- Training the employees to use the new system.

**Entity/Definition Matrix**

| ENTITY | BUSINESS DEFINITION |
|---|---|
| Employee | A person who is currently on the company's payroll or on company benefits. |
| Emergency Contact | Person to contact incase of an emergency at work concerning the employee in which the contact is associated with. Employee can list a primary and secondary emergency contact individual. |
| Job Description | Gives a description of the type of job the employee holds. |
| Activity Code | Gives information about the current status of the employee with the company. |
| Room | A location in the building where one or more employee works. |
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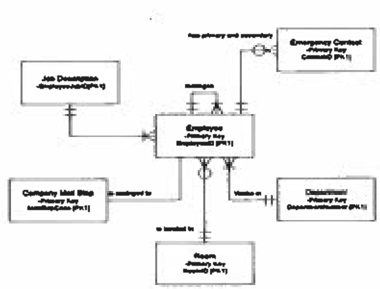
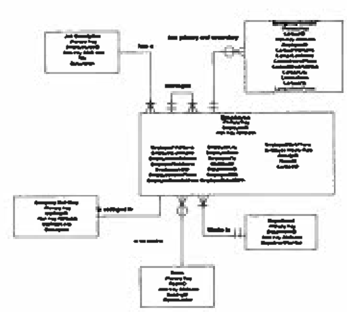
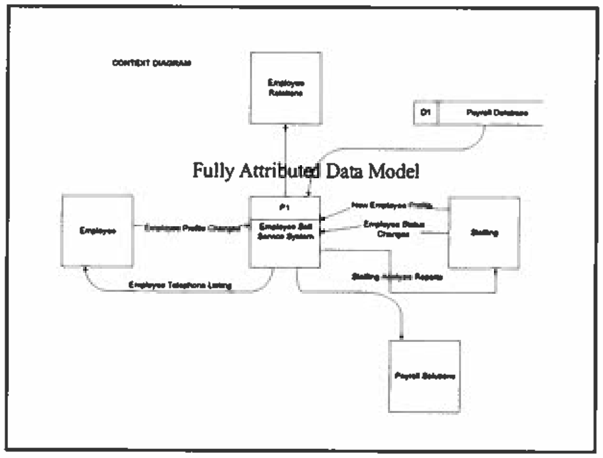
| Department | A specialized unit of employees that work to accomplish the same goal. |
| | |
| | |



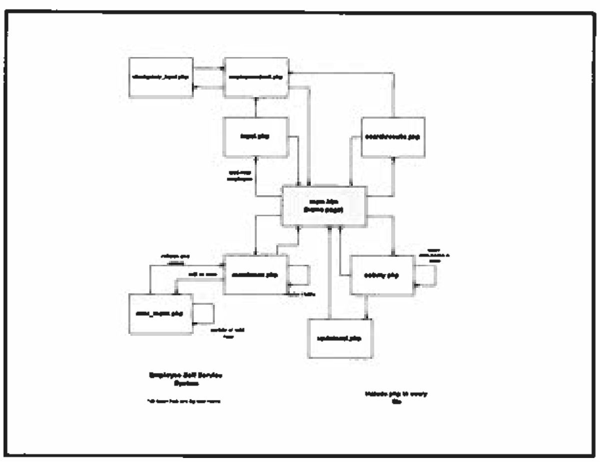
Key-Based Data Model

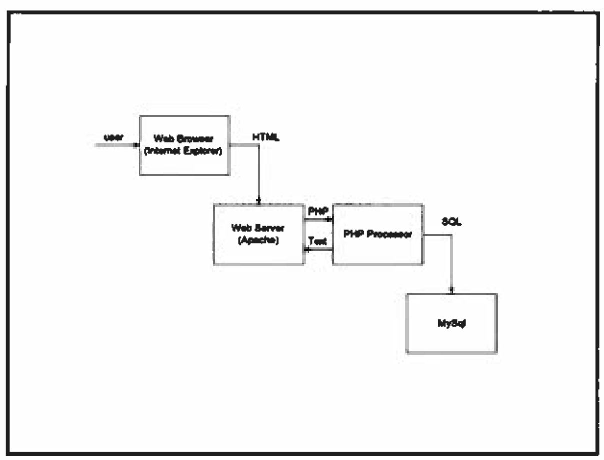


Fully Attributed Data Model




| Characteristics | Candidate 1 | Candidate 2 | Candidate 3 |
|---|---|---|---|
| Portion of System Computerized | COTS package PeopleSoft would be purchased and customized to satisfy [illegible] required functionality. | Employee Information (profile only), United Way and Savings Bonds deductions. | Employee Information(profile only). |
| Benefits | This solution can be put into place quickly because it's a purchased solution. (Some business processes might need to be changed to support software processes.) | Completely supports user required business processes for AI - Information Systems. | Same as Candidate 2. |
| Servers and Workstations | Refer to corporate information architecture for AI information system. | Same as Candidate 1. | A single UNIX box running Apache and MySQL. |
| Software Tools Needed | MS Visual C++ 6.0 and JAVA to customize COTS package. MS Internet Explorer 5.0 Oracle 8.0 | Oracle Forms MS Visual C++ 6.0 System Architect MS Internet Explorer 5.0 Oracle 8.0 | PHP Visio MS Internet Explorer Apache TextEditor MySQL |
| Application Software | Package Human Resource system solution from PeopleSoft Software. | Custom in-house solution. | Same as Candidate 2. |
| Method of Data Processing | Client/Server with Internet interface. | Same as Candidate 1. | Same as Candidate 1. |
| Output Devices and Implications | HP [illegible] Laser Printers connected to the LAN. | Same as Candidate 1. | Same as Candidate 1. |
| Input Devices and Implications | Keyboard and Mouse. | Same as Candidate 1. | Same as Candidate 1. |
| Storage Devices and Implications | Oracle 8.0 Database Management System with 2GB storage capacity. (One for each site.) | Same as Candidate 1. | Unix server with a large disk. |

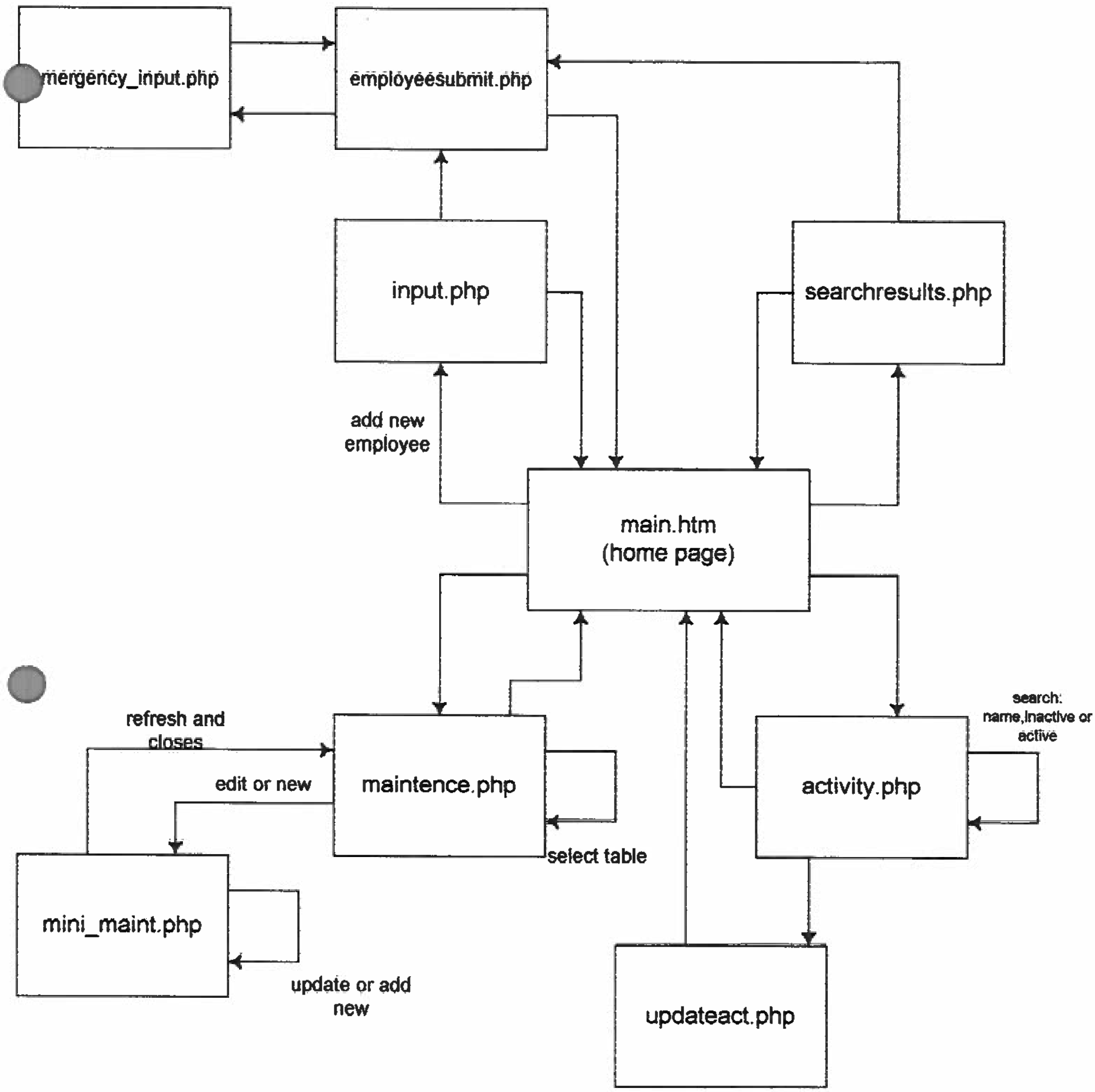




## Coming Attractions

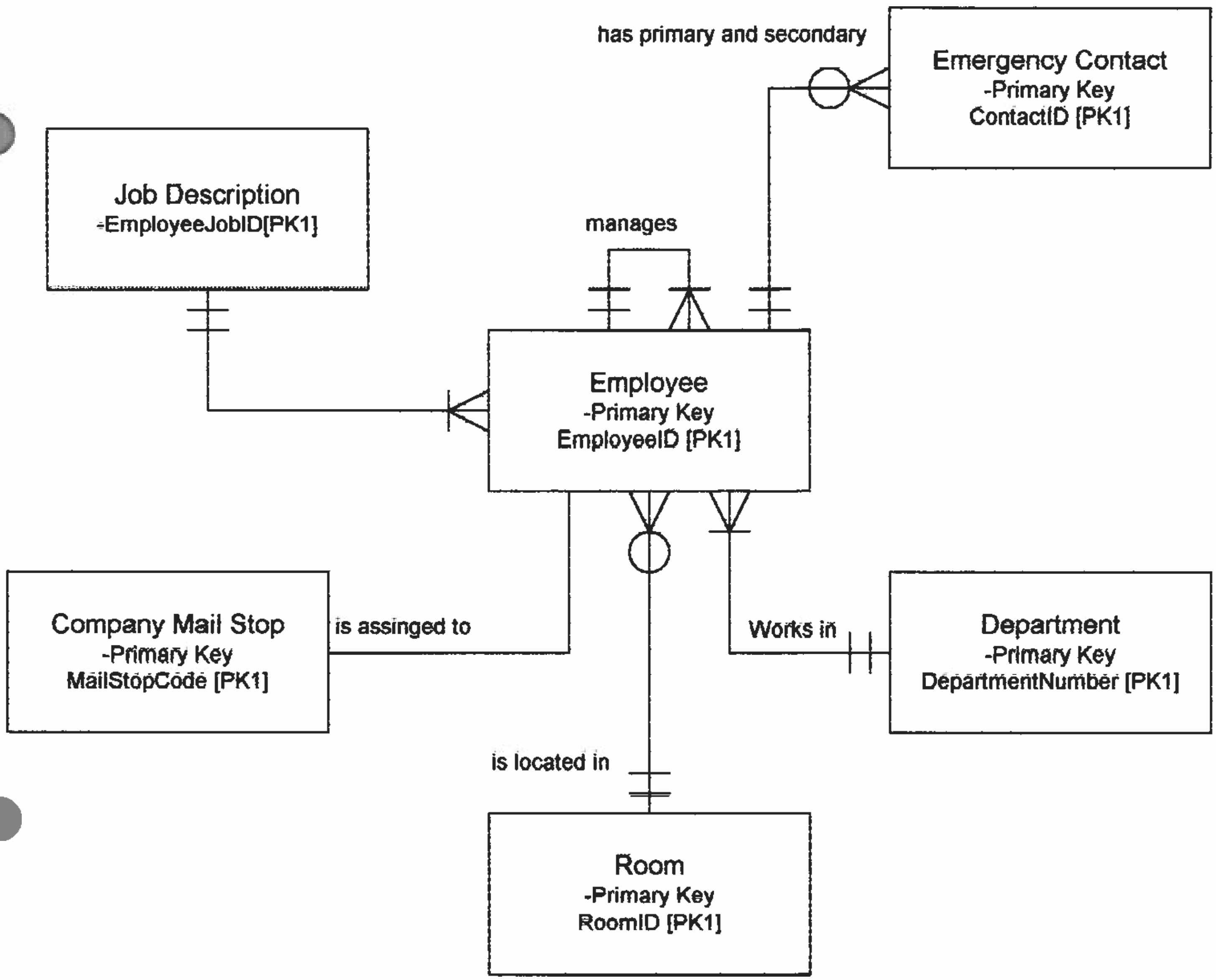
- Security
- Ability to edit or delete emergency contacts
- Ability to edit or delete employees

## Any Questions?




**Employee Self Service System**

*all searches are by last name

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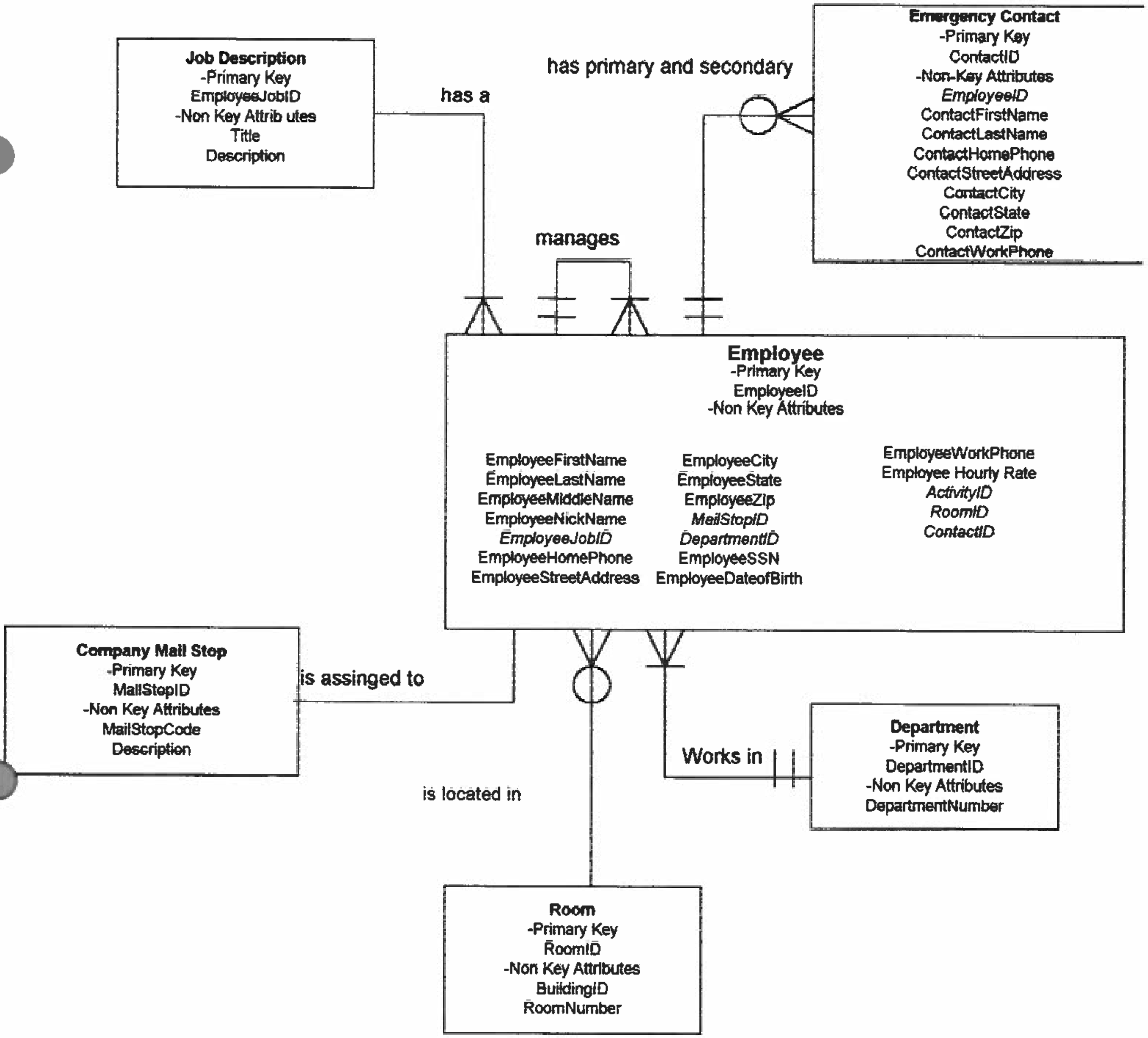


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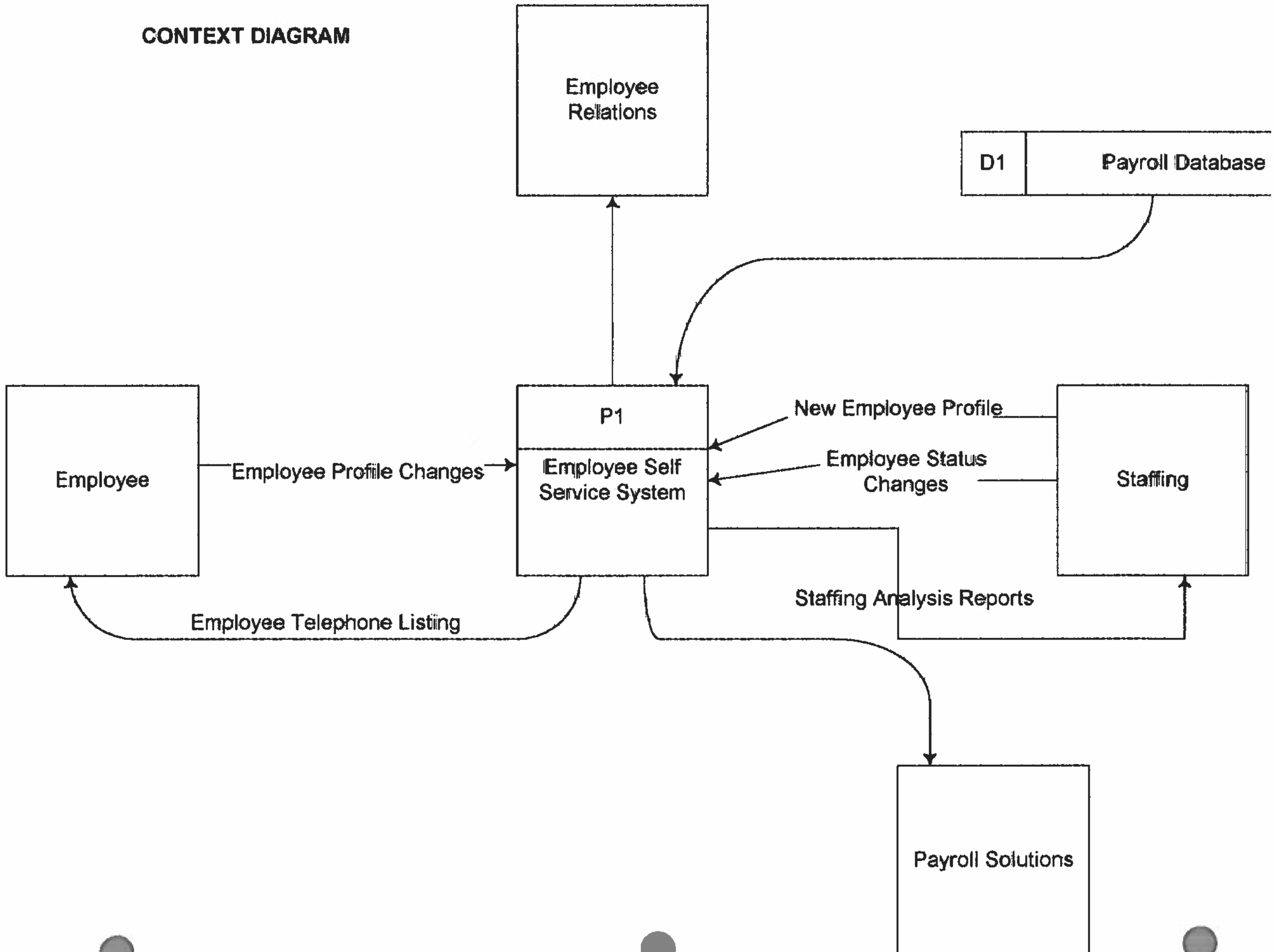
**CONTEXT DIAGRAM**



4-24-2002

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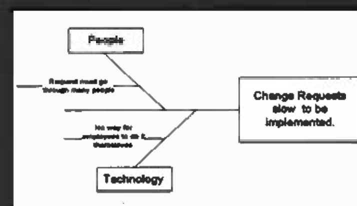


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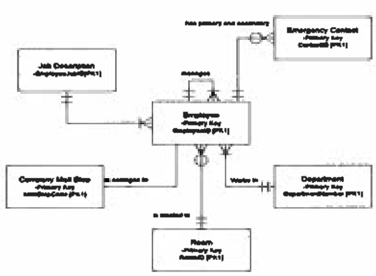


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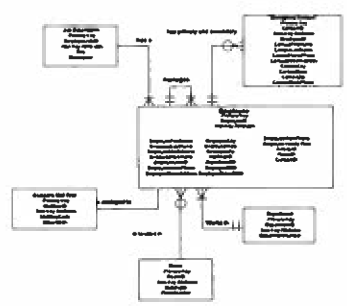

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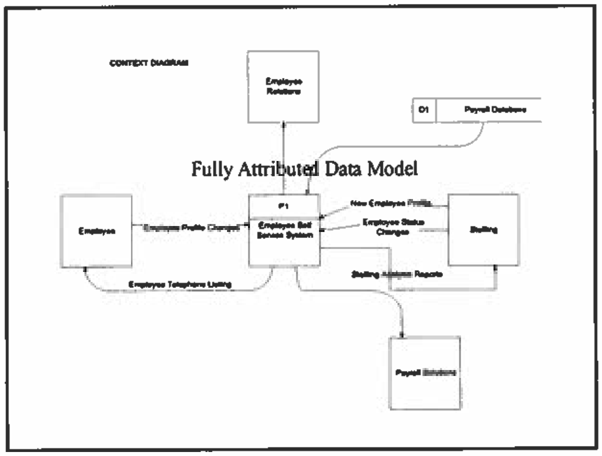


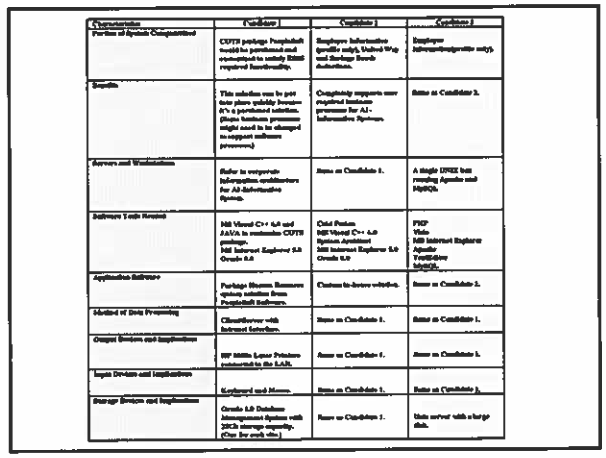

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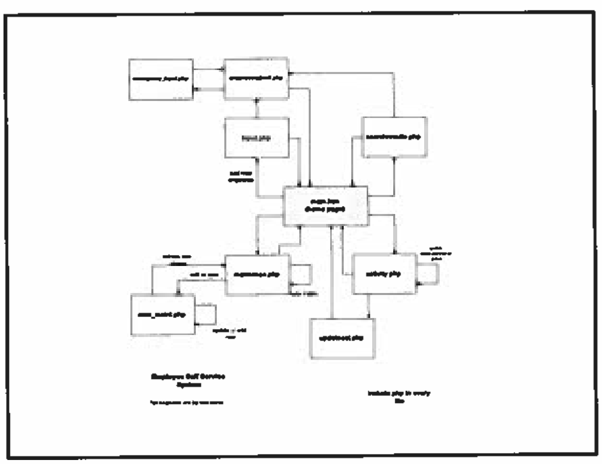


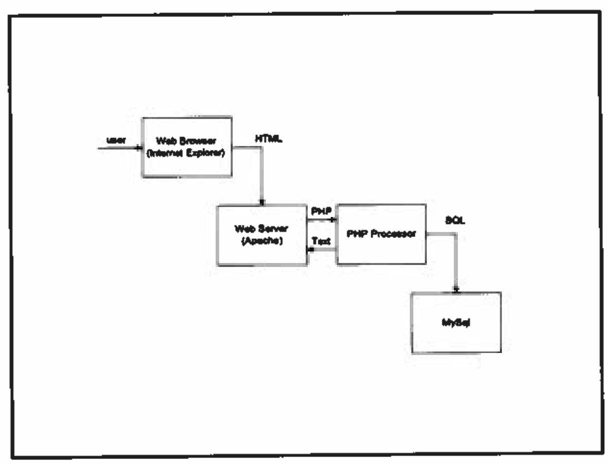


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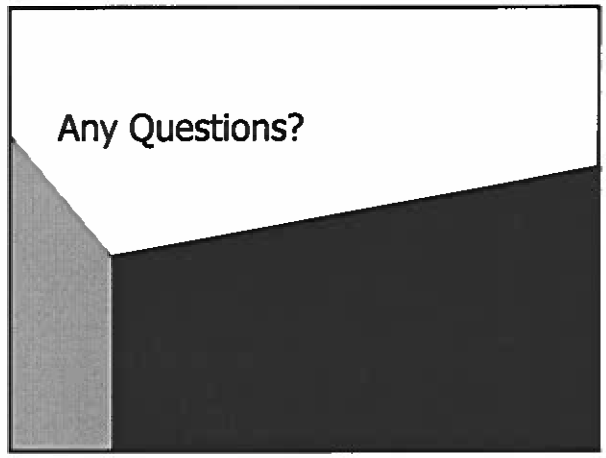
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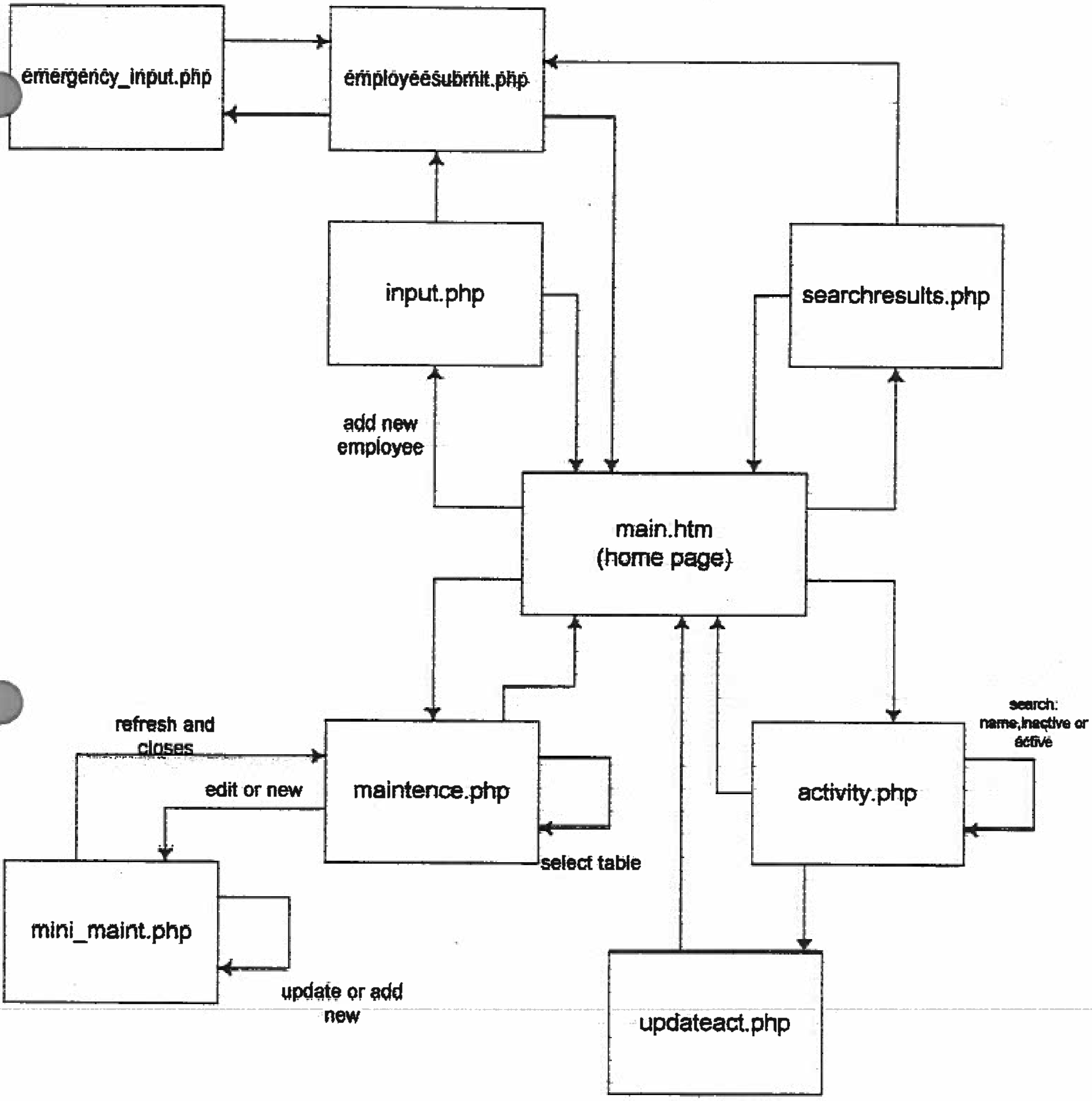


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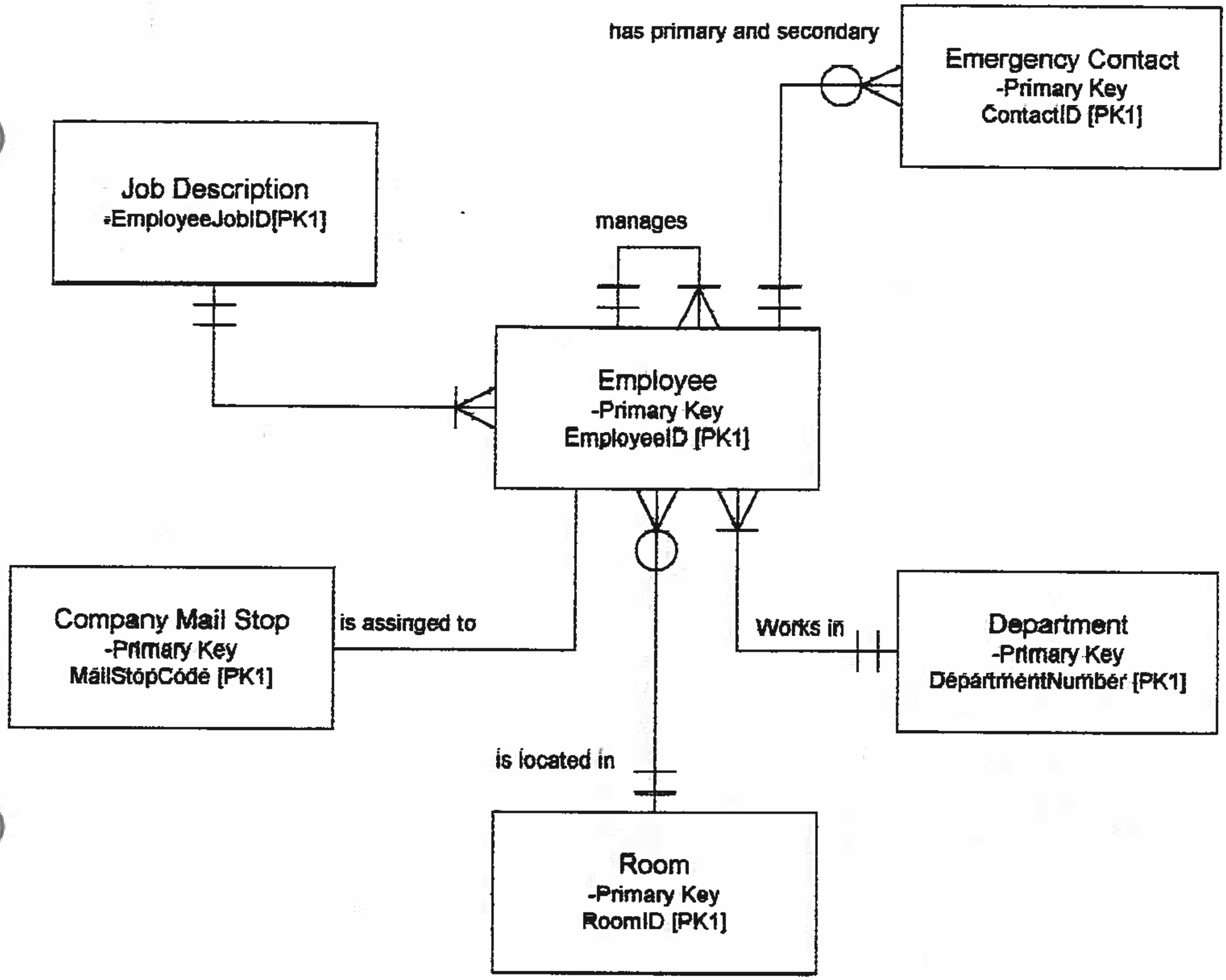


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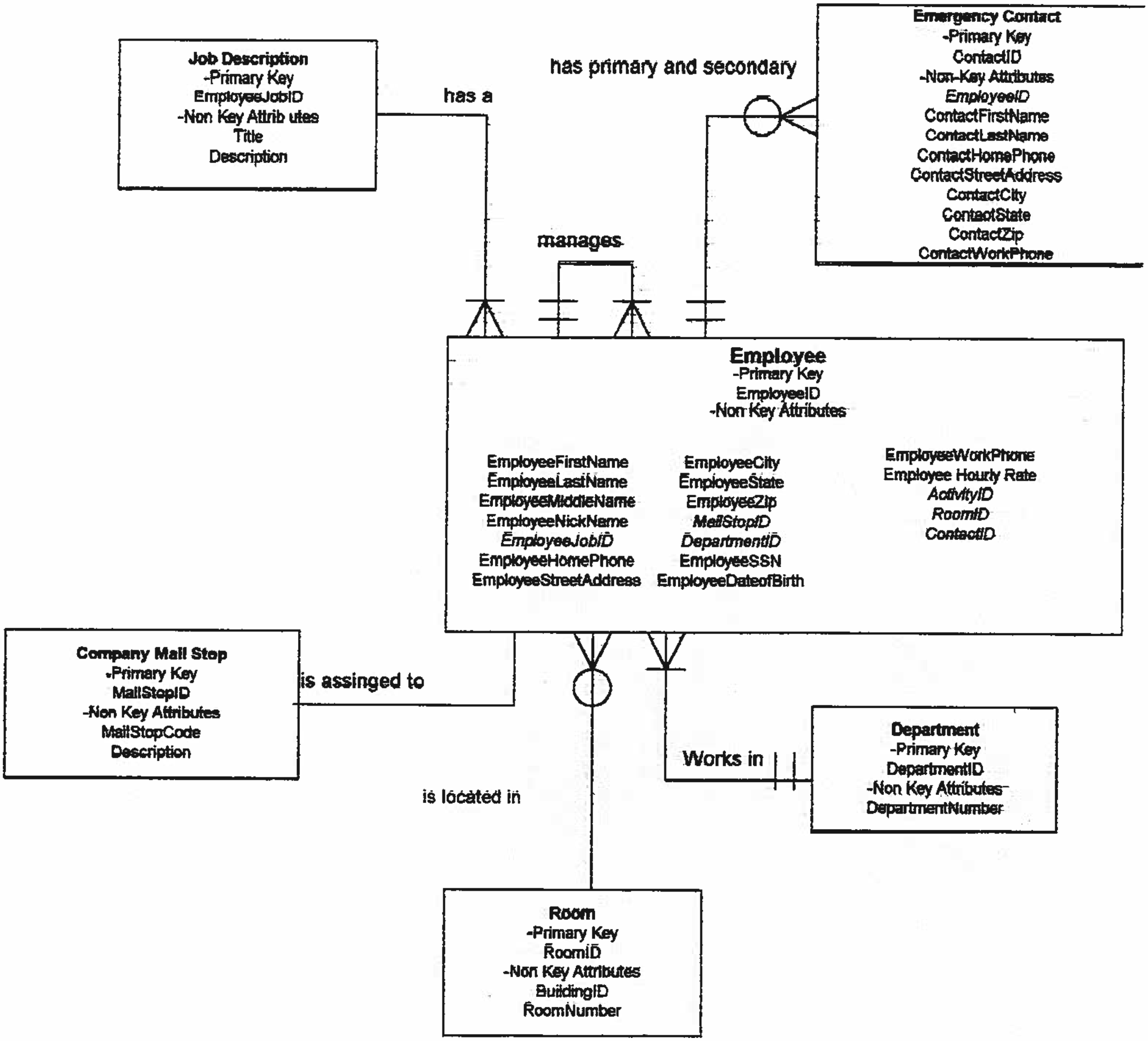


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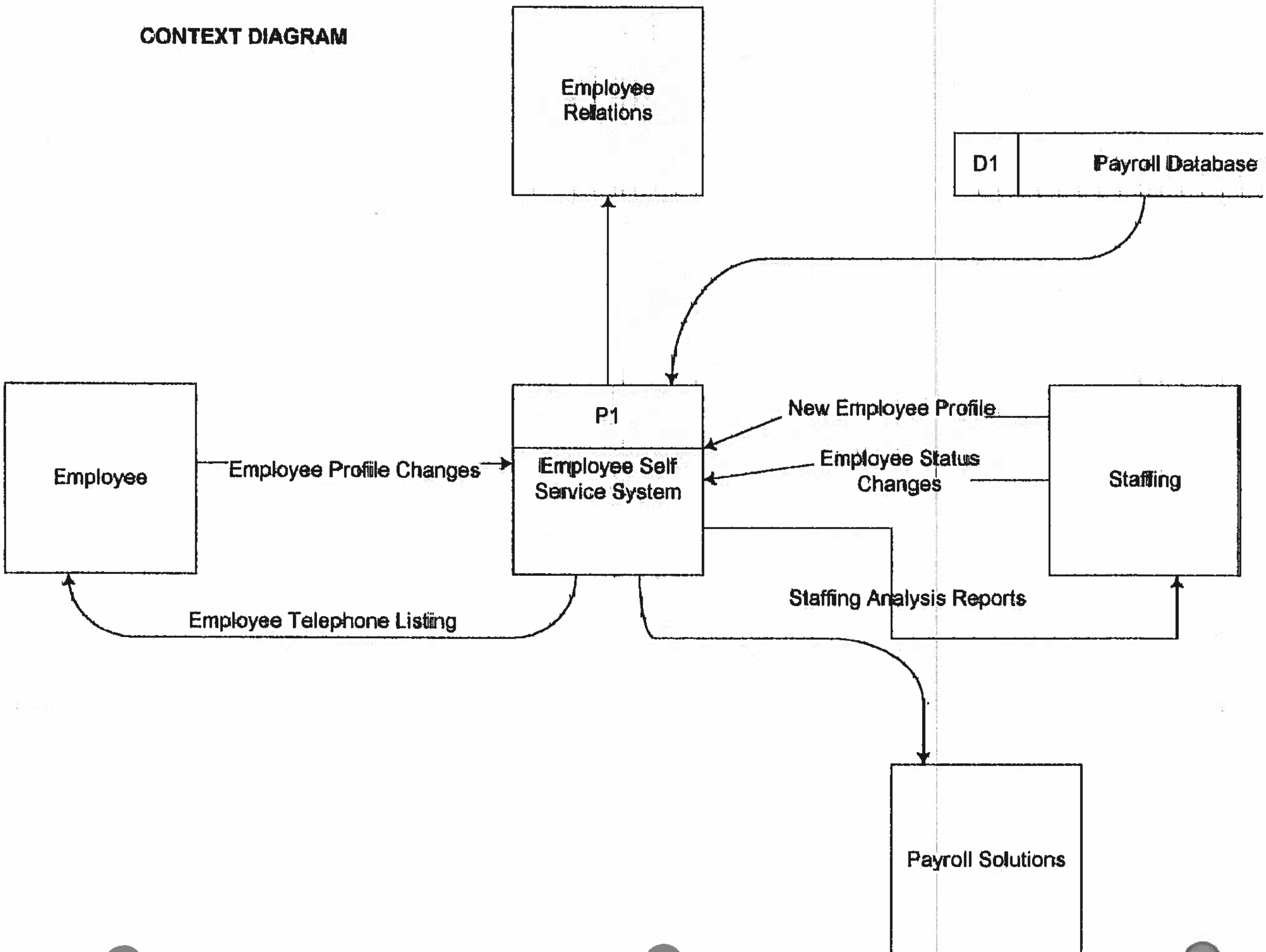

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# Employee Self Service System

## Table of Contents:

T. Coker & J. Varnadore

Phone: (843) 824-5519
Email: saxman101@bigfoot.com,
johnvcsu@hotmail.com

# REQUEST FOR INFORMATION SYSTEM SERVICES

| DATE OF REQUEST | SERVICE REQUESTED FOR DEPARTMENT(S) |
|---|---|
| 1/11/02 | Human Resources |

| SUBMITTED BY (key user contact) | | EXECUTIVE SPONSOR (funding authority) |
|---|---|---|
| Name | J. Briner | Name |
| Title | Doctor in Computer Science | Title |
| Office | Ashby Hall | Office |
| Phone | 863-7529 | Phone |

**TYPE OF SERVICE REQUESTED:**

- ☐ Information Strategy Planning
- ☐ Business Process Analysis and Redesign
- ☒ New Application Development
- ☐ Other (please specify ____________________
- ☐ Existing Application Enhancement
- ☐ Existing Application Maintenance (problem fix)
- ☐ Not Sure

**BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE** (attach additional documentation as necessary)

The current method of maintaining employee information, a request is sent to the human resources department where it must be updated manually. This is a time consuming and expensive process and is prone to error. The company also produces a physical phonebook monthly. As soon as the phone book is printed, it is out of date. The dynamic nature of the company contributes to the work required to keep the system updated. The costs associated with producing the phone book are further exacerbated since the system used to produce the phone book is separate from the main system for maintaining the employee data.

**BRIEF STATEMENT OF EXPECTED SOLUTION**

Employees will maintain their own personal information (address, phone number, beneficiary, etc.). The system should be able to provide an online phone book.

**ACTION (ISS Office Use Only)**

☐ **Feasibility assessment approved** — **Assigned to** T. Coker & J. Varnadore

☐ **Feasibility assessment waived** — **Approved Budget $**
**Start Date** 1/11/02 **Deadline** 4/24/02

☐ **Request delayed** — **Backlogged until date:** ____________

☐ **Request rejected** — **Reason:** ____________________

**Authorized Signatures:**

____________________ ____________________
**Project Executive Sponsor**

*project began in COINS 495 with group members: E. Jenkins, J. Schmizze, K. Larue

# PROBLEM STATEMENT MATRIX

| | | | |
|---|---|---|---|
| **PROJECT:** | **Employee Self-Service System** | **PROJECT MANAGER:** | **J. Briner** |
| **CREATED BY:** | **T. Coker & J. Varnadore** | **LAST UPDATED BY:** | **T. Coker & J. Varnadore** |
| **DATE CREATED:** | **1/28/02** | **DATE LAST UPDATED:** | **4/22/02** |

| Brief Statements of Problem, Opportunity, or Directive | Urgency | Visibility | Annual Benefits | Priority or Rank | Proposed Solution |
|---|---|---|---|---|---|
| 1. Telephone book is soon out of date after printing | 3 months | High | Will Allow Updated Info Immediately | 2 | New Development |
| 2. Employee information maintained in microcomputer and mainframe. | 3 months | Medium | Accessible by all employees | 5 | New development |
| 3. Maintenance of mainframe expensive. | 3 months | Low | Unidentifiable | 3 | New development |
| 4. Employee information is often inconsistent across company systems. | 3 months | High | Unidentifiable | 1 | New development |
| 5. Redundant information in various systems. | 6 months | Medium | Unidentifiable | 6 | Future version of newly developed system. |
| 6. Payroll checks non-deliverable and mail sent to incorrect addresses. | 3 months | Medium | Unidentifiable | 1 | New development |
| 7. Employees unable to locate co-workers in an efficient manner | 3 months | High | Unidentifiable | 2 | New development |
| 8. Reports not delivered in an acceptable time frame. | 3 months | Medium | Unidentifiable | 4 | New development |
| | | | | | |
| | | | | | |

# Project Feasibility Assessment Report

For

# Employee Self Service System

By

Tim Coker & John Varnadore

## PURPOSE

This *Project Feasibility Assessment Report* presents the preliminary findings of our investigation into the feasibility of your project to alleviate concerns about effeciencies of maintaining employee information. This report also outlines our proposal of the procedures and schedule to be followed during this project. If you find any discrepancies or misconceptions, please bring them to our immediate attention.

## PRELIMINARY FINDINGS AND ANALYSIS

After a brief, preliminary investigation of the system being studied, we offer the following observations and initial analysis. The current system has become difficult and expensive to maintain. A completely new system appears to be the best course of action.

### *3*Project Description

This section of the report describes the history that led to this project and the proposed project scope.

**History Leading to this Project Proposal.** The current system is a combination of a mainframe system and a microcomputer. Employees send requests to Human Resources to update their information via paper based forms. The HR employees must enter the data on the mainframe and microcomputer systems. This process is expensive and is prone to error. The mainframe is used for administrative type duties (payroll, etc.) while the microcomputer is used solely to produce a phone book. This phone book is almost out of date as soon as it is printed. Given the current level of growth experienced by the company, keeping data up to date has become overwhelming.

**Scope of this Project.** In the coming weeks, we will be carefully analyzing the project's scope to define a reasonable target and schedule. In the meantime, our preliminary definition is as follows: The main business area that will be affected by this system will be the Human Resources, though all employees will ultimately use the system. It will completely replace the existing mainframe and microcomputer systems and will effectively offload the work of maintaining the employees' information to the employees themselves. It will provide an interface where employees can easily change the address, phone number and other personal information that the company has on file from any location and/or workstation.

The project will address the following business functions:

1. Human Resources

   Maintaining employee Information

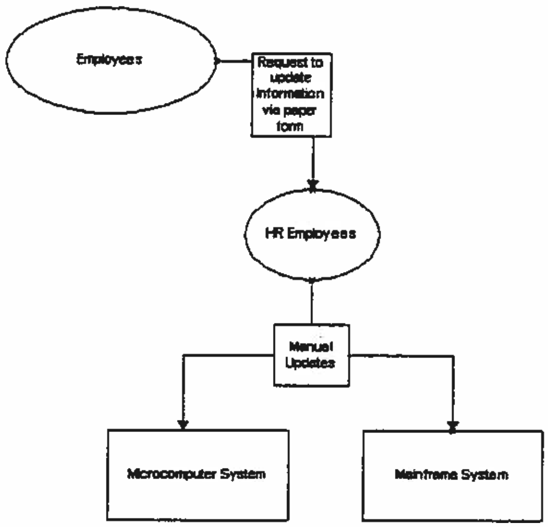
2. All Employees

   Maintaining their personal information

**Project Environment**

This section of the survey describes the project environment in terms of project participants, problems and opportunities to be addressed, and project constraints that will or may limit eventual solutions. ***Figure 1*** represents a context diagram of the current system.

***Figure 1:***




The employees enter submit a request to update their information via a paper employee. HR employees must then enter the data twice in the mainframe and microcomputer systems.

**Project Participants.** To date, we have identified the following list of participants for this project. Please inform us of any potential omissions.

1. Management - Direct Users or Managers of the System

2. Non-Management - Direct Users of the System

3. Other People or Departments Affected by, Interested in, or Interfacing to the System

**Problems and Opportunities.** We have compiled the following list of problems and opportunities to be addressed in the project. The list is not final. In the coming weeks, we will modify the list and provide you with a detailed analysis of problems, opportunities, and solutions. At any time during this project, you should feel free to add to, subtract from, or expand upon this preliminary list.

1. Data must be entered multiple times across multiple systems, creating errors.
2. The mainframe system is expensive to maintain.
3. The phone book is out of date as soon as its published.
4. Updates require an unacceptable amount of time.
5. Paychecks and company mailings are sent to the wrong addresses.

**Project Constraints**. Project constraints are limitations, good or bad, that will or may constrain any solutions that we might propose. Constraints can be technical, monetary, time, or political. To date, we have identified the following preliminary list:

1. The time frame is three months.
2. Training the employees to use the new system.

Our project approach will eventually examine numerous alternative system solutions, and it would be premature to commit to any solution at this time. However, it is never too early to begin brainstorming and cataloging ideas.

**Client's perceptions of what they want or expect.** It is our understanding that you envision a new or improved systems that allows employees to update their information themselves through a standard interface. It will also provide an electronic phone book accessable to all employees.

**The analyst's perceptions of possible solutions and ideas.** This system will be a web-based system with a combination of PHP or ASP and Oracle or mySQL-Server for the backend.

## PROPOSAL

We propose to write a system that will allow users to update their personal information through the corporate intranet. This section of the report outlines my proposal.

### Project Schedule Overview

See the attached Gantt chart for the project schedule.

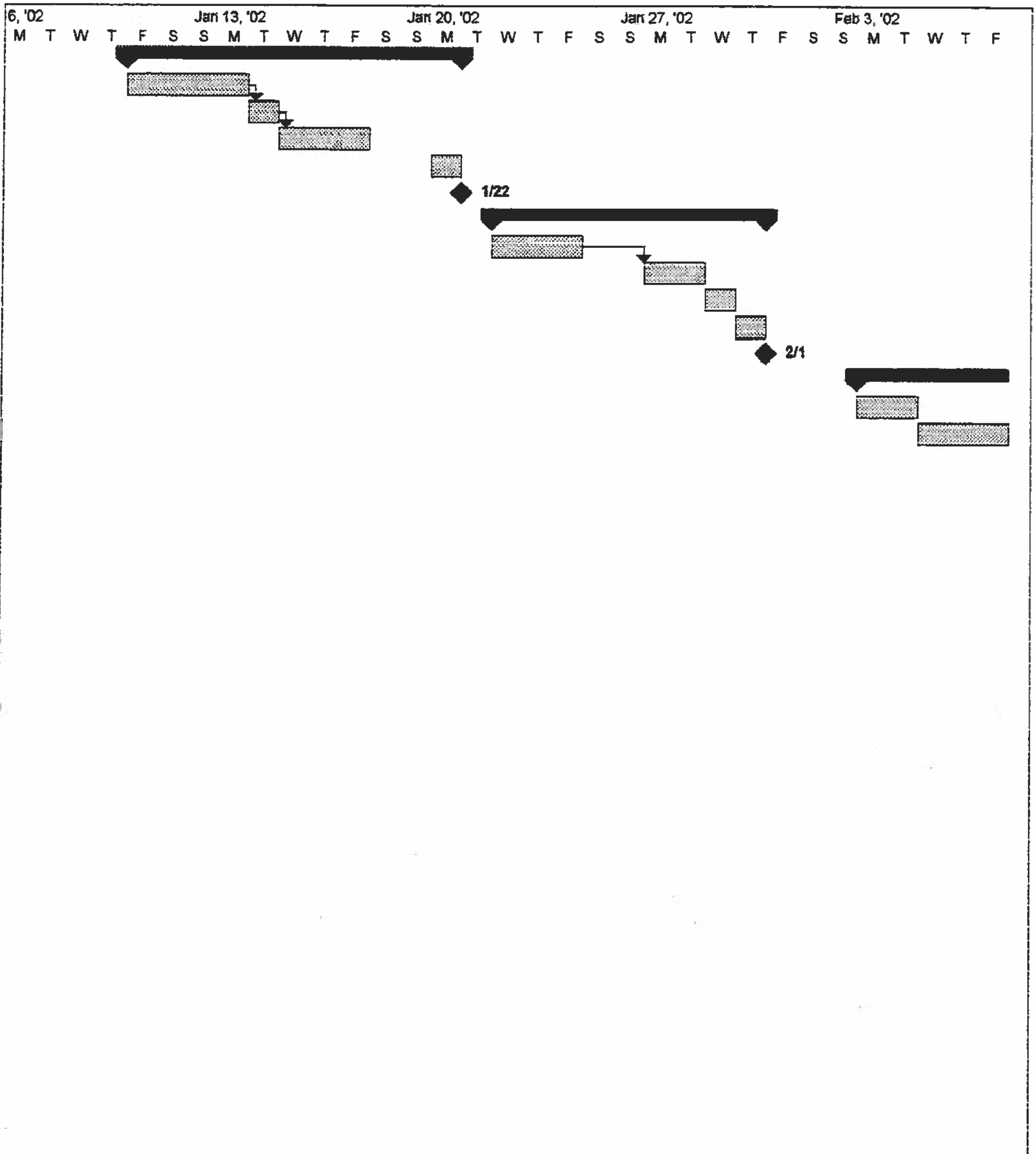
| ID | | Task Name | Duration | Start |
|---|---|---|---|---|
| 1 | | 1 Prelimary Investigation | 7 days | Fri 1/11/02 |
| 2 | | 1.1 List problems, opportunties, and directives | 2 days | Fri 1/11/02 |
| 3 | | 1.2 Negotiate scope | 1 day | Tue 1/15/02 |
| 4 | | 1.3 Plan the project | 3 days | Wed 1/16/02 |
| 5 | | 1.4 Present the project and plan | 1 day | Mon 1/21/02 |
| 6 | | 1.5 Project charter completed | 0 days | Tue 1/22/02 |
| 7 | | 2 Problem Analysis | 7 days | Wed 1/23/02 |
| 8 | | 2.1 Analyze the current system | 3 days | Wed 1/23/02 |
| 9 | | 2.2 Establish system improvement objectives | 2 days | Mon 1/28/02 |
| 10 | | 2.3 Update the project plan | 1 day | Wed 1/30/02 |
| 11 | | 2.4 Present Findings and recommendations | 1 day | Thu 1/31/02 |
| 12 | | 2.5 Problem Statement completed | 0 days | Fri 2/1/02 |
| 13 | | 3 Requirements Analysis | 8 days | Mon 2/4/02 |
| 14 | | 3.1 Identify business requirements | 2 days | Mon 2/4/02 |
| 15 | | 3.2 Analysis system requirements | 3 days | Wed 2/6/02 |
| 16 | | 3.3 Prioritize business requirements | 2 days | Mon 2/11/02 |
| 17 | | 3.4 Update the project plan | 1 day | Wed 2/13/02 |
| 18 | | 3.5 Requirements statement completed | 0 days | Wed 2/13/02 |
| 19 | | 4 Decision Analysis | 12 days | Thu 2/14/02 |
| 20 | | 4.1 Identify candidate solutions | 3 days | Thu 2/14/02 |
| 21 | | 4.2 Analyze candidate solutions | 3 days | Tue 2/19/02 |
| 22 | | 4.3 Recommend a target solution | 3 days | Fri 2/22/02 |
| 23 | | 4.4 Recommend a project solution | 3 days | Wed 2/27/02 |
| 24 | | 4.5 System recommendation completed | 0 days | Mon 3/4/02 |
| 25 | | 5 Design | 14 days | Tue 3/5/02 |
| 26 | | 5.1 Design the application architecture | 4 days | Tue 3/5/02 |
| 27 | | 5.2 Design the system database | 2 days | Mon 3/11/02 |
| 28 | | 5.3 Design the system interface | 3 days | Wed 3/13/02 |
| 29 | | 5.4 Design the application logic | 3 days | Mon 3/18/02 |
| 30 | | 5.5 Update the project plan | 2 days | Thu 3/21/02 |
| 31 | | 6 Construction | 10 days | Mon 3/25/02 |
| 32 | | 6.1 Build and test networks | 2 days | Mon 3/25/02 |
| 33 | | 6.2 Build and test databases | 3 days | Wed 3/27/02 |
| 34 | | 6.3 Install and test new software packages | 2 days | Mon 4/1/02 |
| 35 | | 6.4 Write and test new programs | 3 days | Wed 4/3/02 |
| 36 | | 7 Implementation | 11 days | Mon 4/8/02 |
| 37 | | 7.1 Conduct system tests | 2 days | Mon 4/8/02 |
| 38 | | 7.2 Prepare Conversion Plan | 1 day | Wed 4/10/02 |
| 39 | | 7.3 Install Databases | 2 days | Thu 4/11/02 |
| 40 | | 7.4 Train system users | 4 days | Mon 4/15/02 |
| 41 | | 7.5 Convert to New system | 2 days | Fri 4/19/02 |

Sep 23, '01

W T F S S M T W T F S

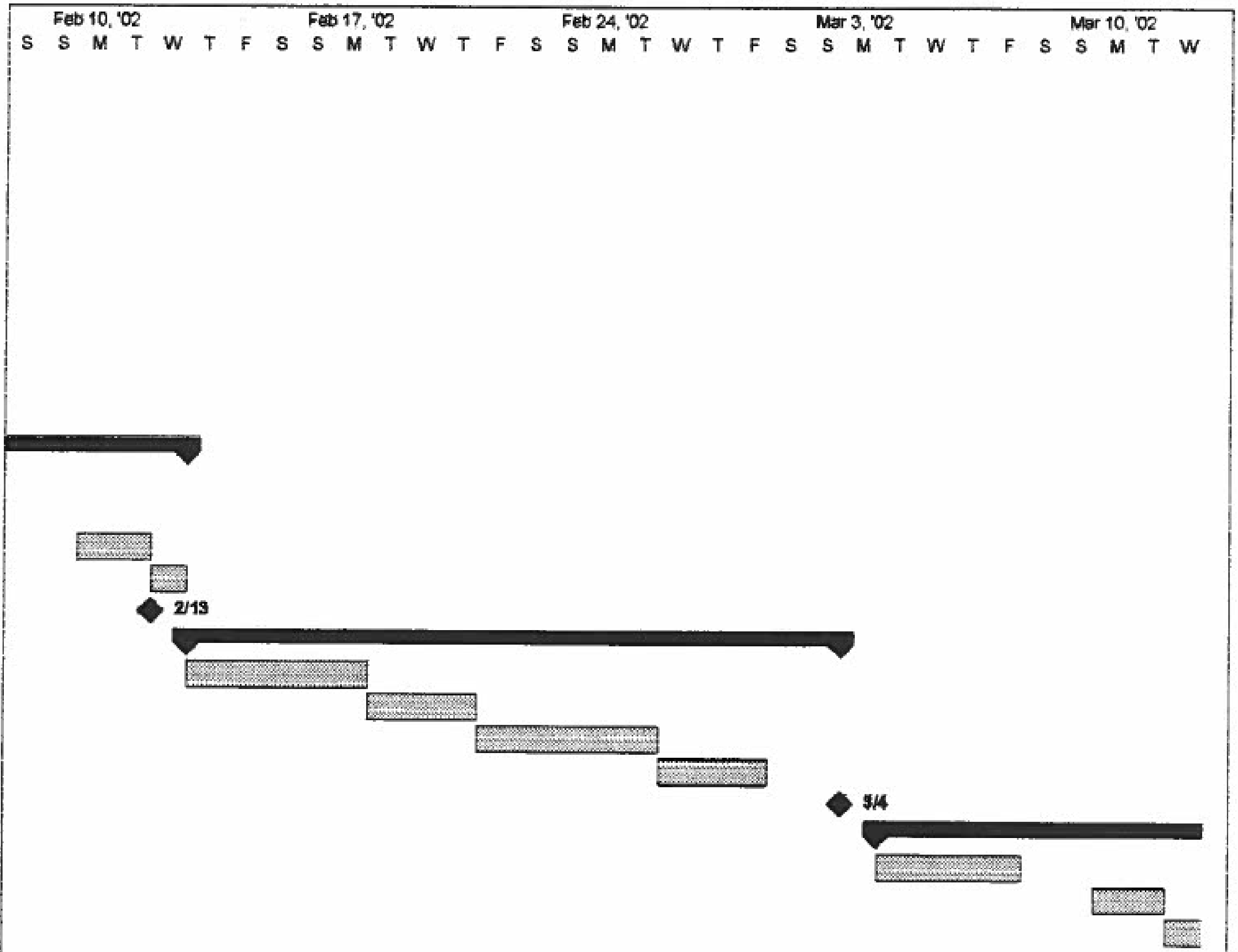
Project: Project1
Date: Thu 4/18/02

Task
Split
Progress
Milestone
Summary
Project Summary
External Tasks
External Milestone
Deadline



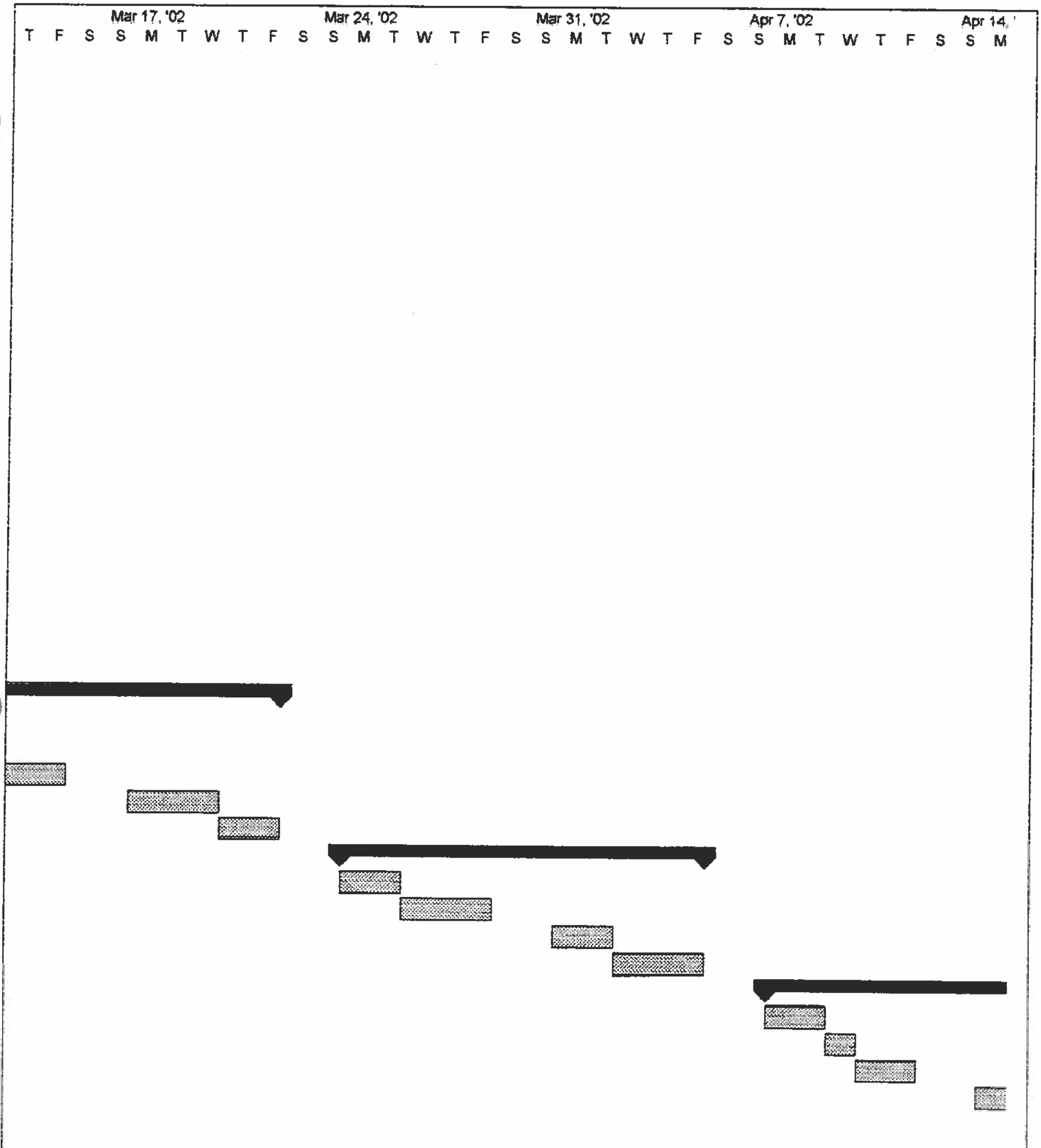

Project: Project1
Date: Thu 4/18/02

Task
Split
Progress
Milestone
Summary
Project Summary
External Tasks
External Milestone
Deadline



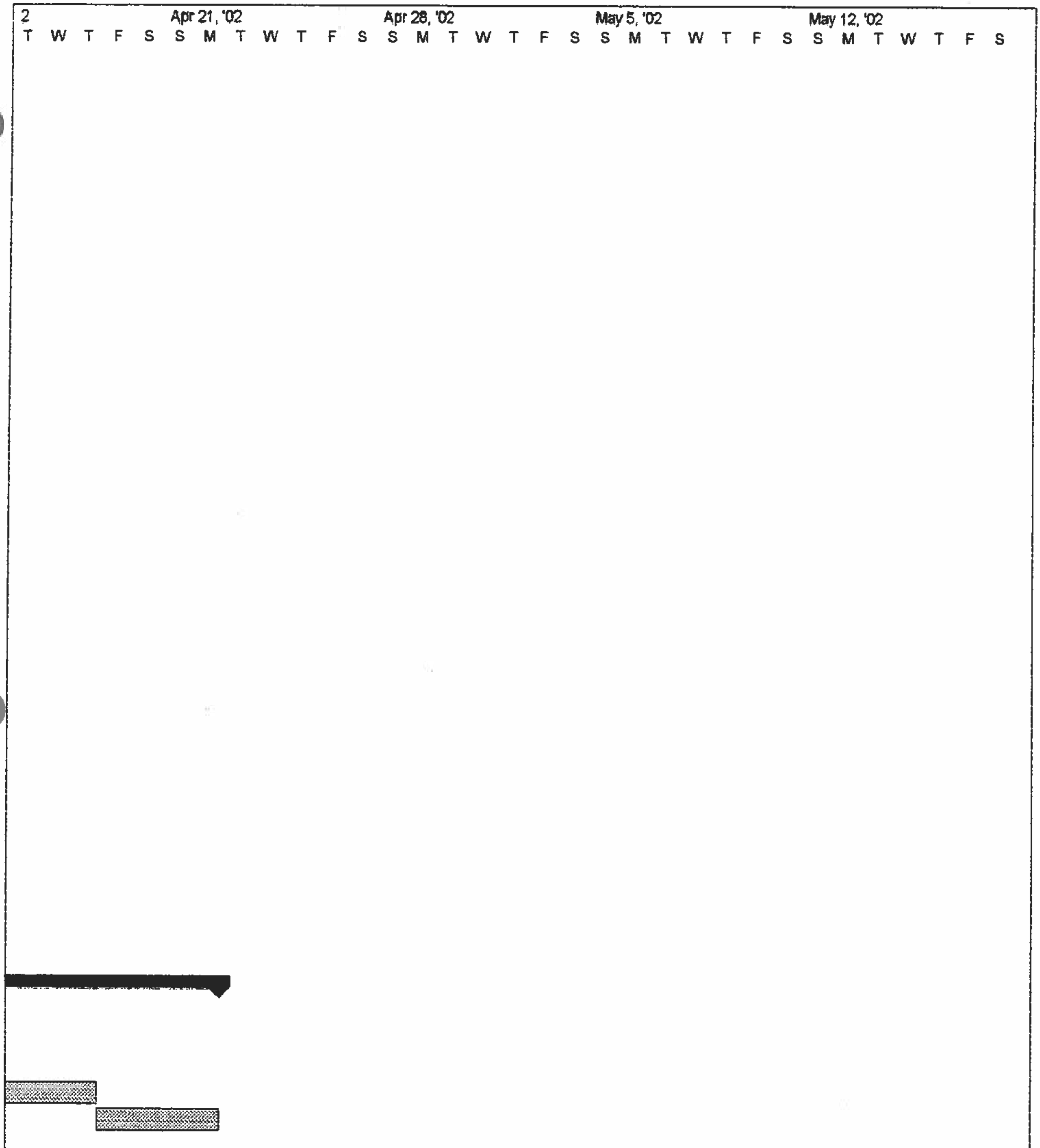

Project: Project1
Date: Thu 4/18/02

Task
Split
Progress
Milestone
Summary
Project Summary
External Tasks
External Milestone
Deadline




Project: Project1
Date: Thu 4/18/02

| | | | |
|---|---|---|---|
| Task | | Project Summary | |
| Split | | External Tasks | |
| Progress | | External Milestone | |
| Milestone | | Deadline | |
| Summary | | | |




Project: Project1
Date: Thu 4/18/02

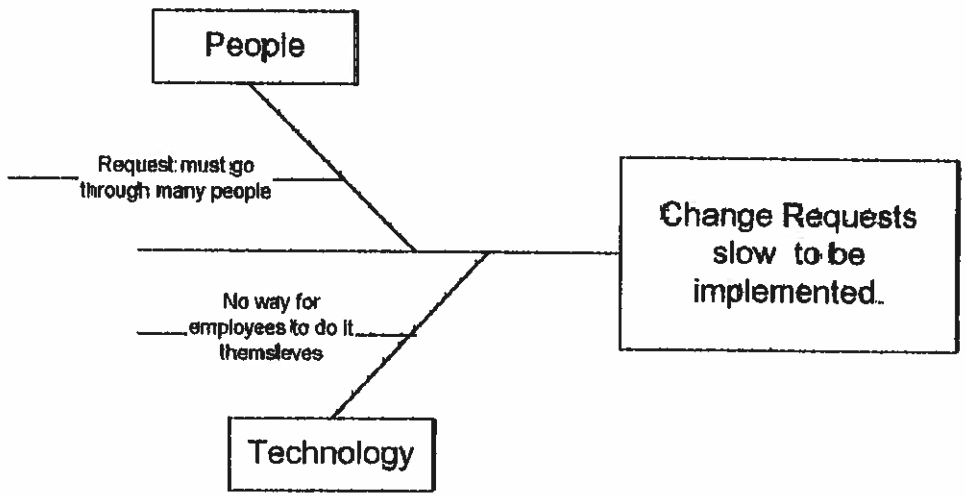
| | | | |
|---|---|---|---|
| Task | | Project Summary | |
| Split | | External Tasks | |
| Progress | | External Milestone | |
| Milestone | | Deadline | |
| Summary | | | |

# PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

| | | | |
|---|---|---|---|
| **Project:** | **Employee Self Service System** | **Project Manager:** | J. Briner |
| **Created by:** | **T. Coker & J. Varnadore** | **Last Updated by:** | **T. Coker & J. Varnadore** |
| **Date Created:** | **1/14/2002** | **Date Last Updated:** | **1/14/2002** |

| CAUSE AND EFFECT ANALYSIS | | SYSTEM IMPROVEMENT OBJECTIVES | |
|---|---|---|---|
| **Problem or Opportunity** | **Causes and Effects** | **System Objective** | **System Constraint** |
| 1. Telephone book in hardcopy form, and it is difficult to update the information. | 1. Personnel changes everyday with this growing company so there are changes in the information daily, so quarterly publication is not sufficient. | 1. Eliminate the Telephone books from being in hard copy form.<br>2. Create a contact list that is online, so it would be easily accessible, to all users.<br>3. Cut down on the time that the administrator has to maintain the system, by allowing the users to update their own information. The administrators only have to add new employees with the ability to update when needed.<br>4. Allow users to input information update themselves in real-time.<br>5. Do away with the need for micro standalone application. | 1. The system must be accessible from remote locations and desktops alike.<br>2. Must be securing so unauthorized data cannot be added to the system. |
| 2. Micro and mainframe both used when possible to maybe just use one when employee information is being | 1. Information about personnel is not stored in the same source repository. | 1. Eliminate the micro system and therefore the duplication of information being keyed into the source. Providing | 1. The system must be accessible from remote locations and desktops alike. |

| | | | |
|---|---|---|---|
| entered into the system. | 2. There isn't an interface between the mainframe and micro applications. | one source of entry for personnel information. | 2 Must be secured so unauthorized data cannot be added to the system. |
| 4. Personnel data maybe outdated or incorrect throughout the entire company systems. | 1. Information has to be keyed in by an administrator, after they have been submitted by the users on forms. This cause a delay before information is actually stored in the system.<br>2. The Information entered is not in real-time, after the administrator keys in the data, they information doesn't actually get out to the users until the new quarterly hardcopy telephone book is published. | 1. Have employees be able to enter and update their personal information themselves.<br>2. Have the old data that is already present be able to interface with the data that is going to be newly entered by the employees. Provide a single source of entry for the employee's information. | 1. New information that is entered into the system must be of same specifications of the old system specifications. |
| 5. Different databases have duplicate employee information. | 1. Employee databases are application specific and not enterprise specific.<br>2. Don't have a single source to store employee information. | 1. Make a single source repository to store the employee information. | |
| 6. Because of the information not being updated probably, the company mailings are incorrect and payroll checks are not delivered. Employee not able to locate other employee's information quickly and sufficiently. | 1. Information has to be keyed in by an administrator, after they have been submitted by the users on forms. This cause a delay before information is actually stored in the system.<br>2. There is not a place that employee information is located that other employees can access their information. | 1. Have real-time changes to information inputted by the employees.<br>2. Provide a single source of entry for the employee's information.<br>3. Have online employee listing accessible to all employees. | 1. System must be platform independent and accessible from desktops.<br>2. Must be secured so unauthorized data cannot be added to the system. |
| 7. Users are unable to access reports in a sufficient time period. | 1. The IS personnel cannot keep up with the request of the employees in a time efficient manner, because of backup of request and other job priorities they are responsible for.<br>2. Current technology doesn't support the ad-hoc report and inquiry functionality. | 1. Provide an ad-hoc reporting and inquiry feature for the users. | 1. Must be secured so unauthorized access to the employee data on the system. |


People
Request must go through many people
No way for employees to do it themsleves
Technology
Change Requests slow to be implemented.

# Study Report

I. Executive summary

A. Summary of recommendation

We suggest that the system be moved to a relational database with a Web interface.

B. Summary of problems, opportunities, and directives

The system is currently expensive to maintain and the data is often not current with other systems. Reports are not available in a timely fashion.

II. Overview of the current system

A. Strategic implications

Other systems using employee information will need to be modified to access the data in the new system.

B. Models of current system

1. Interface model

   The users send a request to the HR dept who then implements the changes among two separate systems. A phone book is printed quarterly.

2. Data model

   Data is stored in the mainframe system and the microcomputer system.

3. Geographic models

The company is spread out over the country and requests must be directed to the data center in Florida for entry by the HR department.

4. Process model

Users fill out change request forms and they are sent to Human Resources where the requested changes are made.

III. Analysis of the current system

A. Performance problems, opportunities, and cause-effect analysis

The requests require a long time to take effect since there are so many stops in the chain. Reports are also unavailable in a timely manner.

B. Information problems, opportunities and cause-effect analysis

The information is maintained in two different systems causing inconsistencies. Consolidating the data in a flexible source will greatly improve the availability of the information.

C. Economic problems, opportunities, and cause-effect analysis

The mainframe is expensive to maintain. Implementing a system by which employees can maintain their own information will enable savings in labor.

D. Efficiency problems, opportunities, and cause-effect analysis

Many hands must touch the data resulting in potential errors and prolonging the time required for an update.

## IV. Detailed recommendations

### A. System improvement objectives and priorities

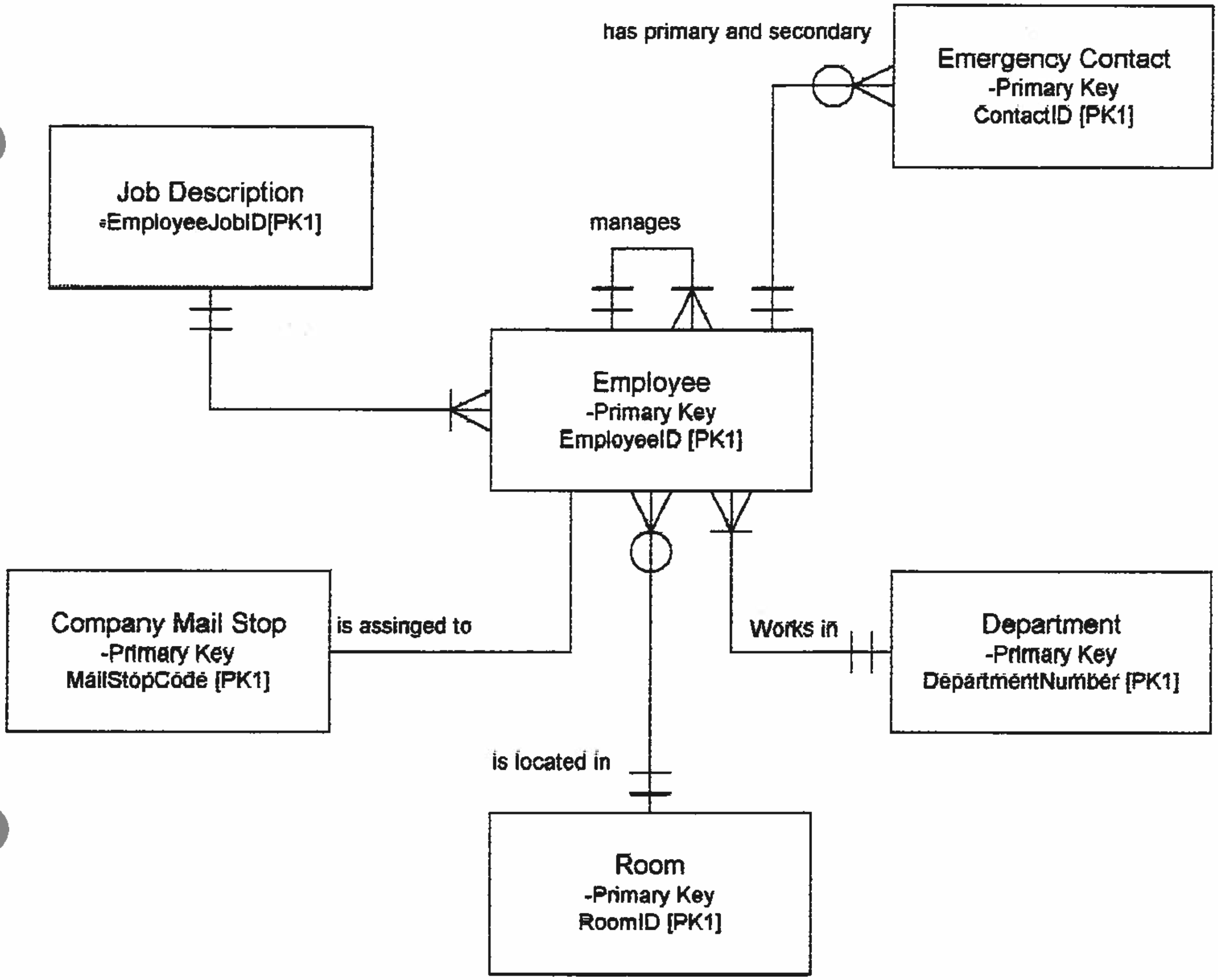
The system will enable employees to update their own information and provide an online phone book.

### B. Constraints

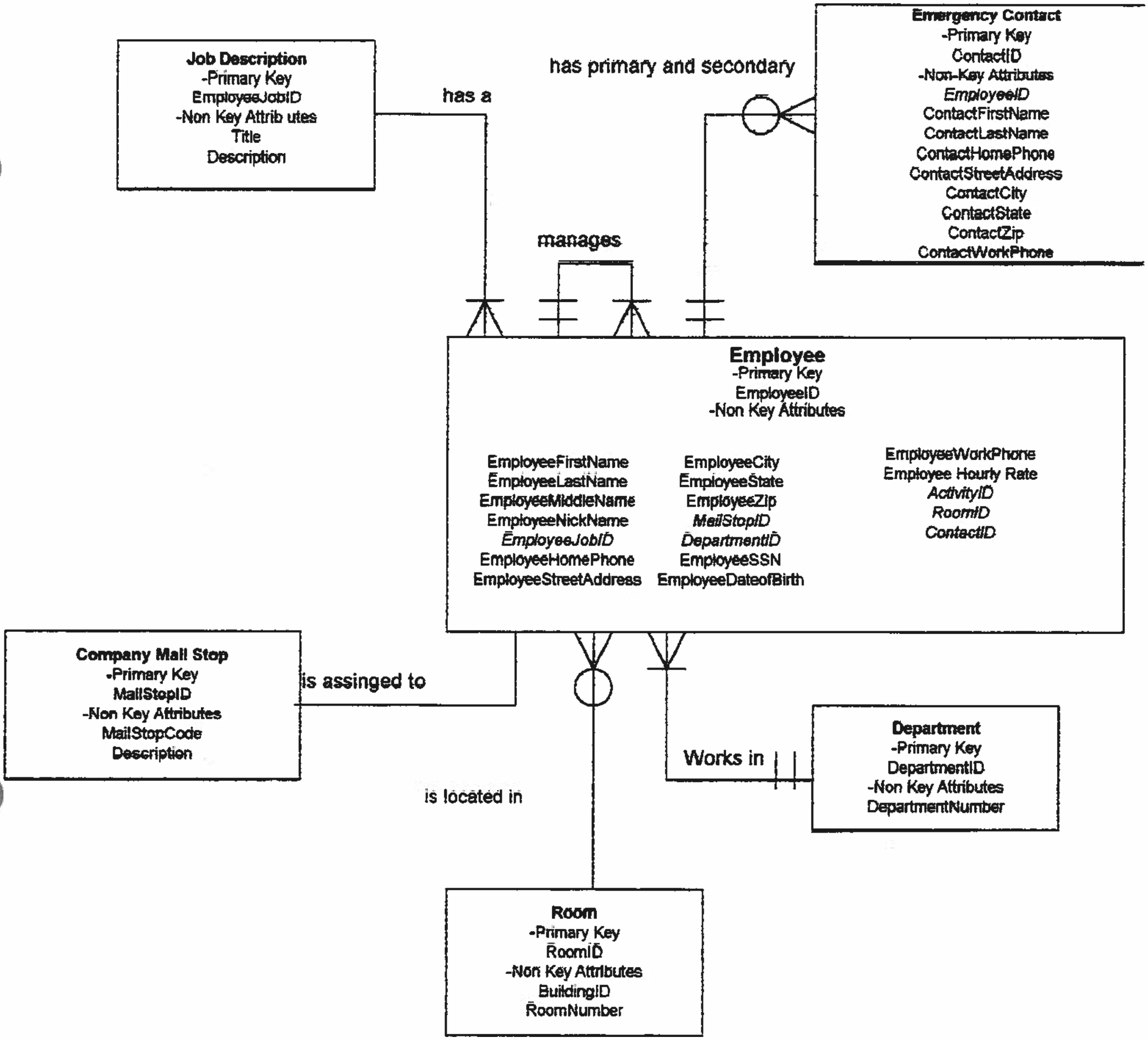
We are limited to a time frame of 6 months and will have to train the employees to use the new system.

**Entity/Definition Matrix**

| ENTITY | BUSINESS DEFINITION |
|---|---|
| **Employee** | A person who is currently on the company's payroll or on company benefits. |
| **Emergency Contact** | Person to contact incase of an emergency at work concerning the employee in which the contact is associated with. Employee can list a primary and secondary emergency contact individual. |
| **Job Description** | Gives a description of the type of job the employee holds. |
| **Activity Code** | Gives information about the current status of the employee with the company. |
| **Room** | A location in the building where one or more employee works. |
| **Company Mail Stop** | Location within the building where employees mail is received and distributed, either inter-company mail or external mail. |
| **Department** | A specialized unit of employees that work to accomplish the same goal. |
| | |
| | |



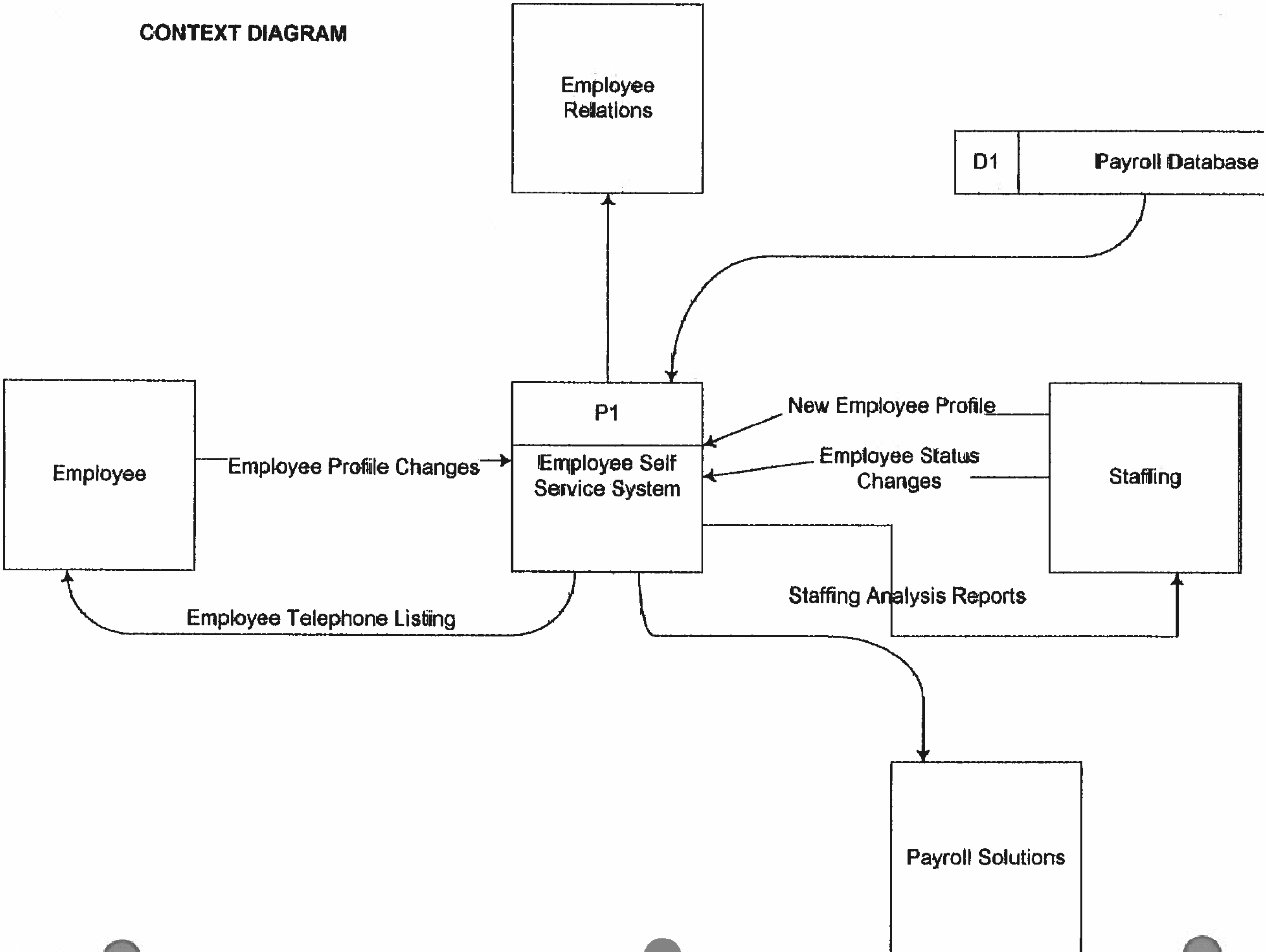

**Key-Based Data Model**



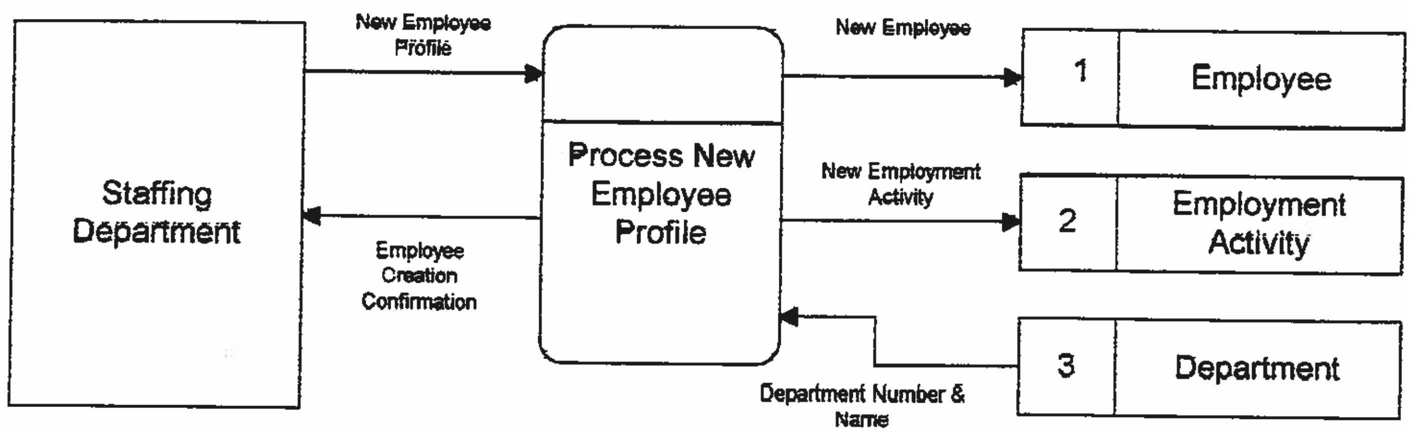

# Fully Attributed Data Model

*Italicized* entries are Foreign Keys

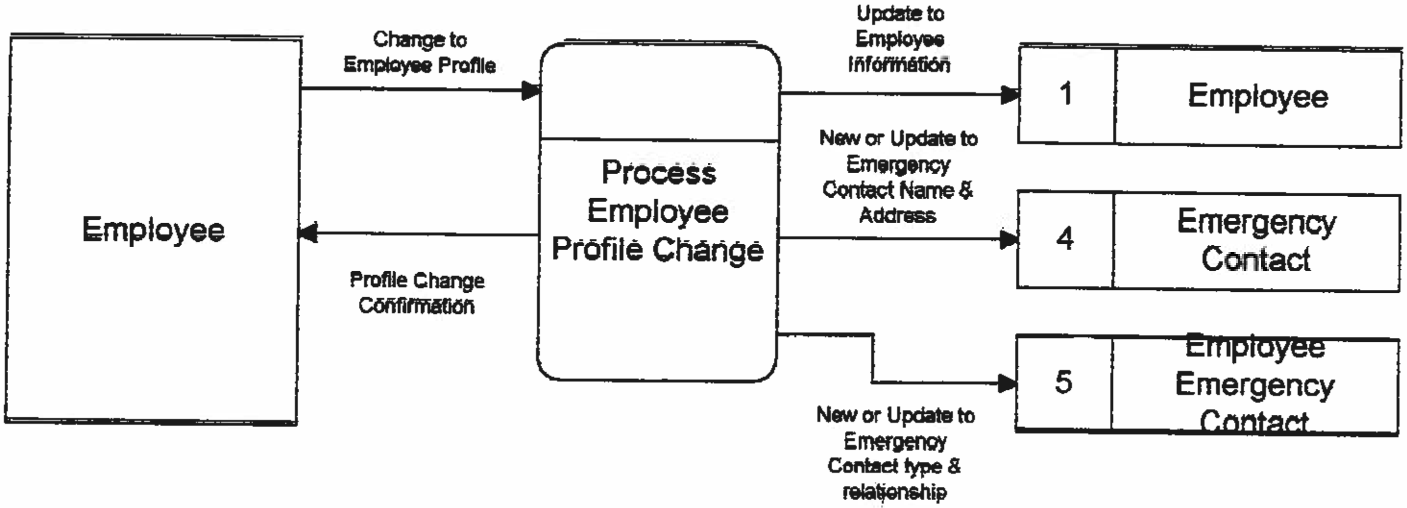
## CONTEXT DIAGRAM



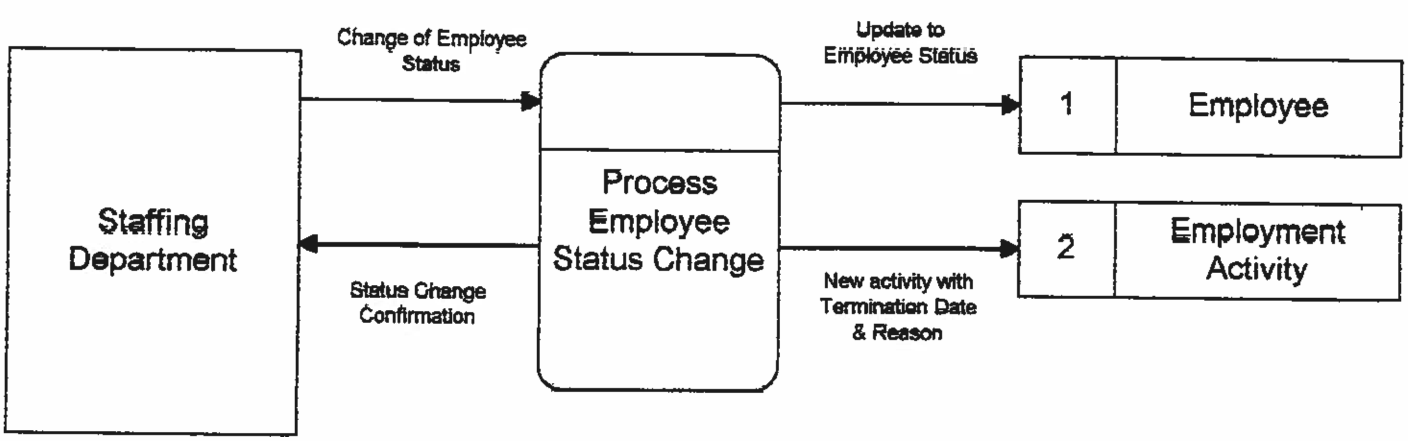
## Event Diagram
## Event:Staffing submits new employee profile



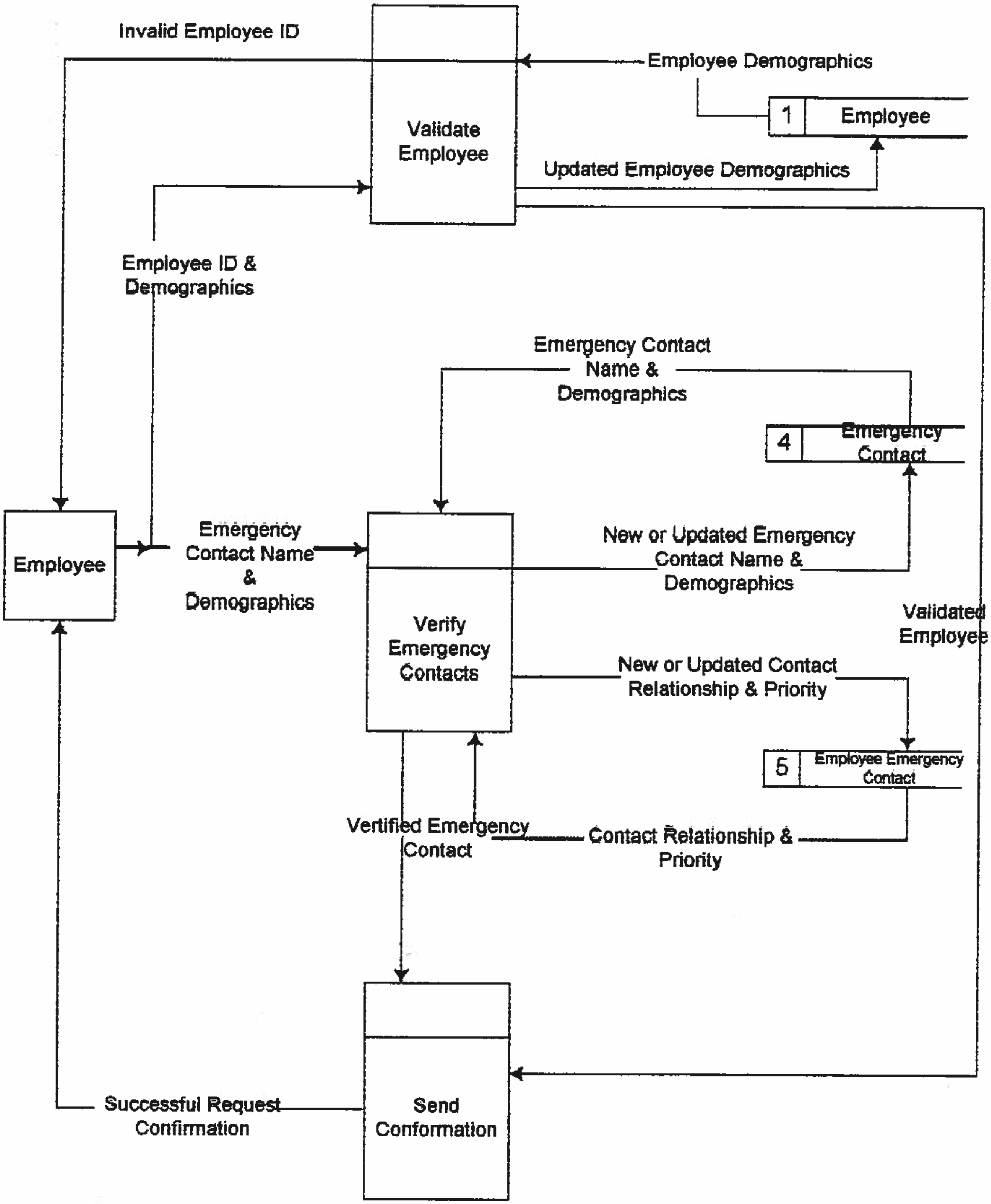

## Event Diagram
## Event:Employee changes profile information



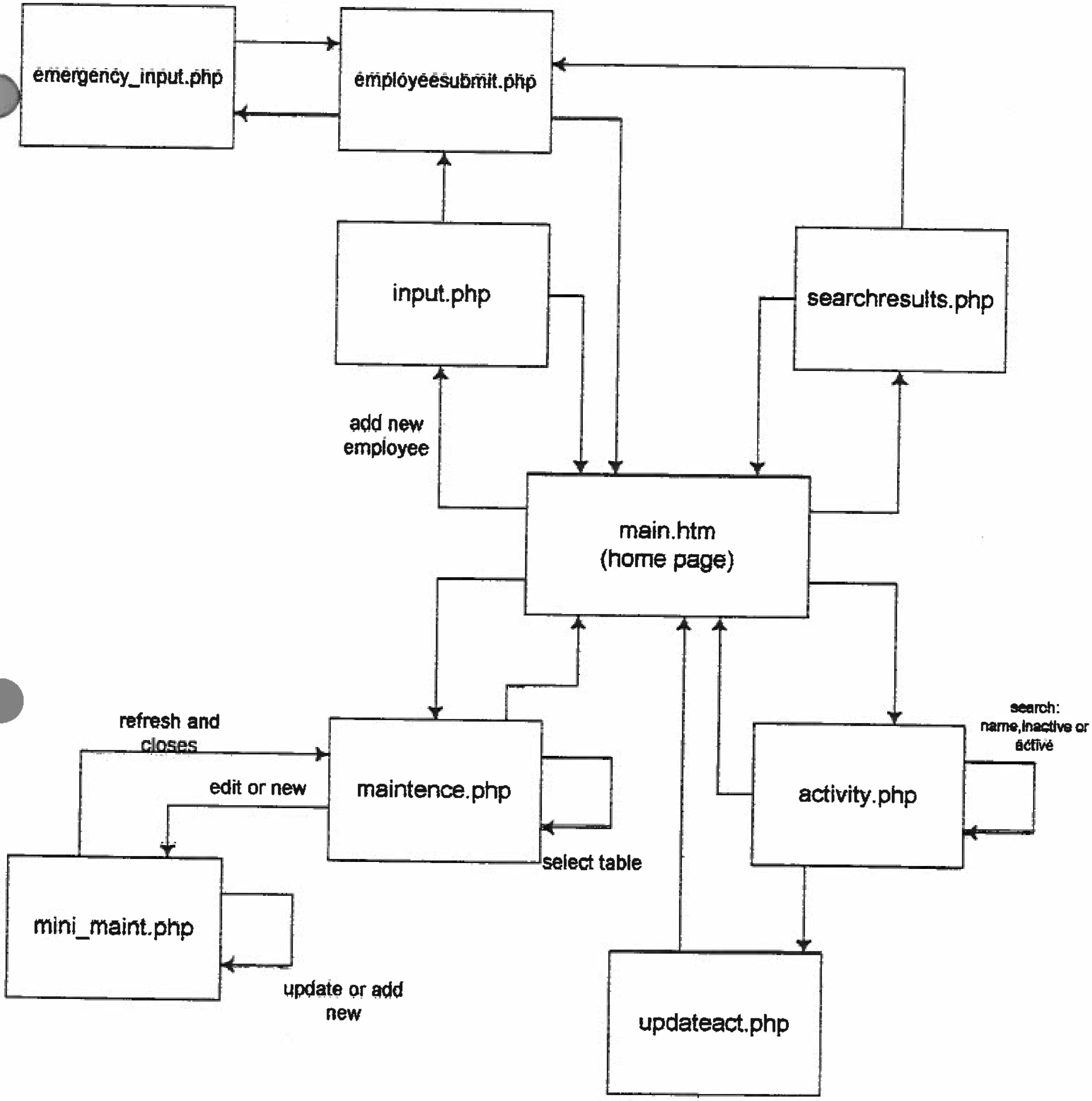

## Event Diagram
## Event:Staffing submits employee status changes



## PRIMITIVE DIAGRAM

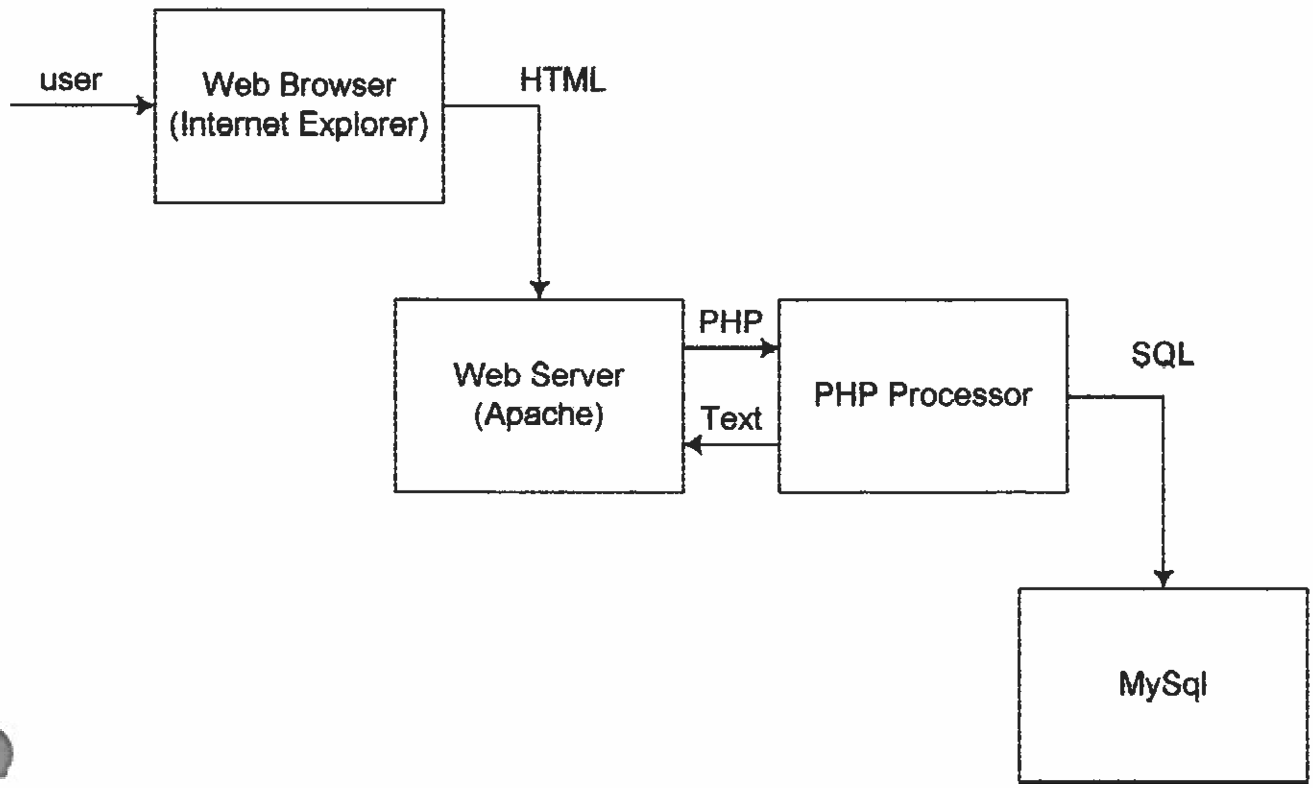


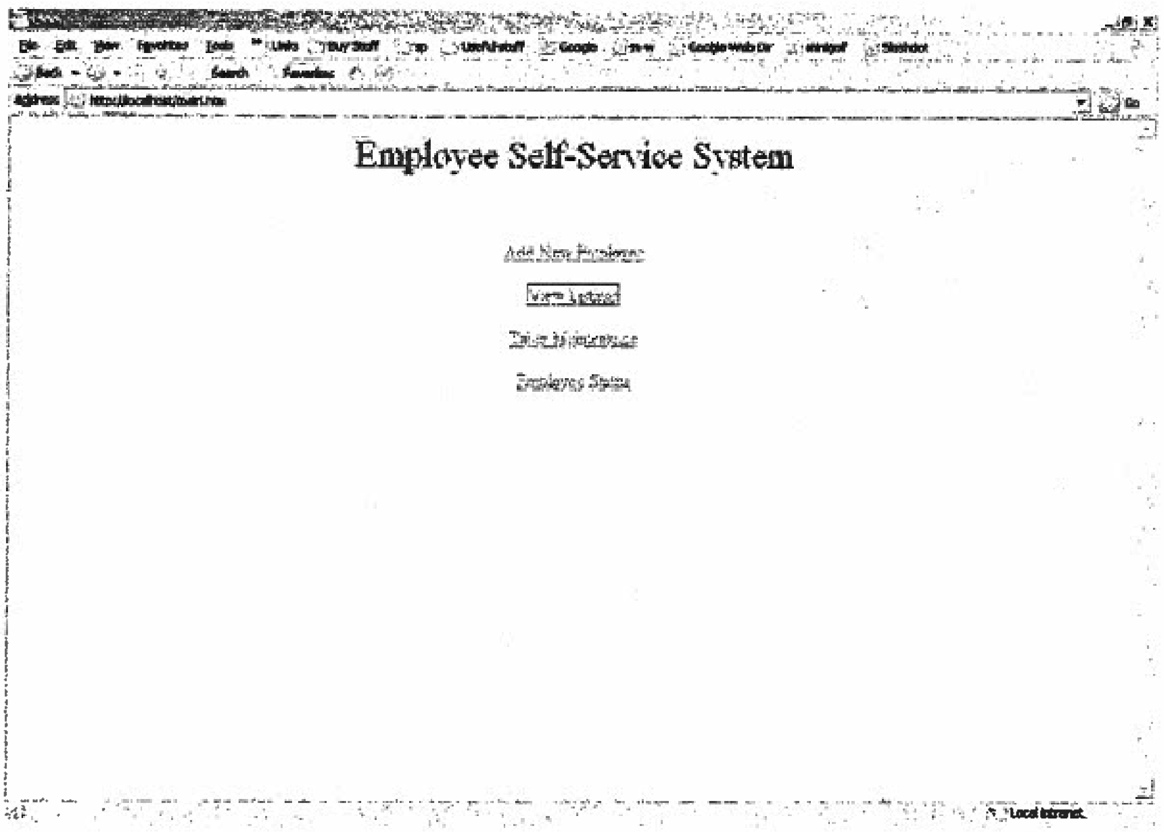
| Characteristics | Candidate 1 | Candidate 2 | Candidate 3 |
|---|---|---|---|
| **Portion of System Computerized**<br><br>Brief description of that portion of the system that would be computerized in this candidate. | COTS package PeopleSoft would be purchased and customized to satisfy ESSS required functionality. | Employee Information (profile only), United Way and Savings Bonds deductions. | Employee Information(profile only). |
| **Benefits**<br><br>Brief description of the business benefits that would be realized for this candidate. | This solution can be put into place quickly because it's a purchased solution. (Some business processes might need to be changed to support software processes.) | Completely supports user required business processes for A1-Information Systems. | Same as Candidate 2. |
| **Servers and Workstations**<br><br>A description of the servers and workstations needed to support this candidate. | Refer to corporate information architecture for A1-Information System. | Same as Candidate 1. | A single UNIX box running Apache and MySQL |
| **Software Tools Needed**<br><br>Software tools needed to design and build the candidate (e.g. database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased. | MS Visual C++ 6.0 and JAVA to customize COTS package.<br>MS Internet Explorer 5.0<br>Oracle 8.0 | Cold Fusion<br>MS Visual C++ 6.0<br>System Architect<br>MS Internet Explorer 5.0<br>Oracle 8.0 | PHP<br>Visio<br>MS Internet Explorer<br>Apache<br>TextEditor<br>MySQL |
| **Application Software**<br><br>A description of the software to be purchased, built, accessed, or some combination of these techniques. | Package Human Resource system solution from PeopleSoft Software. | Custom in-house solution. | Same as Candidate 2. |
| **Method of Data Processing**<br><br>Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time. | Client/Server with Intranet Interface. | Same as Candidate 1. | Same as Candidate 1. |
| **Output Devices and Implications**<br><br>A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints). | HP 8000c Laser Printers connected to the LAN. | Same as Candidate 1. | Same as Candidate 1. |
| **Input Devices and Implications**<br><br>A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs). | Keyboard and Mouse. | Same as Candidate 1. | Same as Candidate 1. |
| **Storage Devices and Implications**<br><br>Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized. | Oracle 8.0 Database Management System with 25Gb storage capacity. (One for each site.) | Same as Candidate 1. | Unix server with a large disk. |

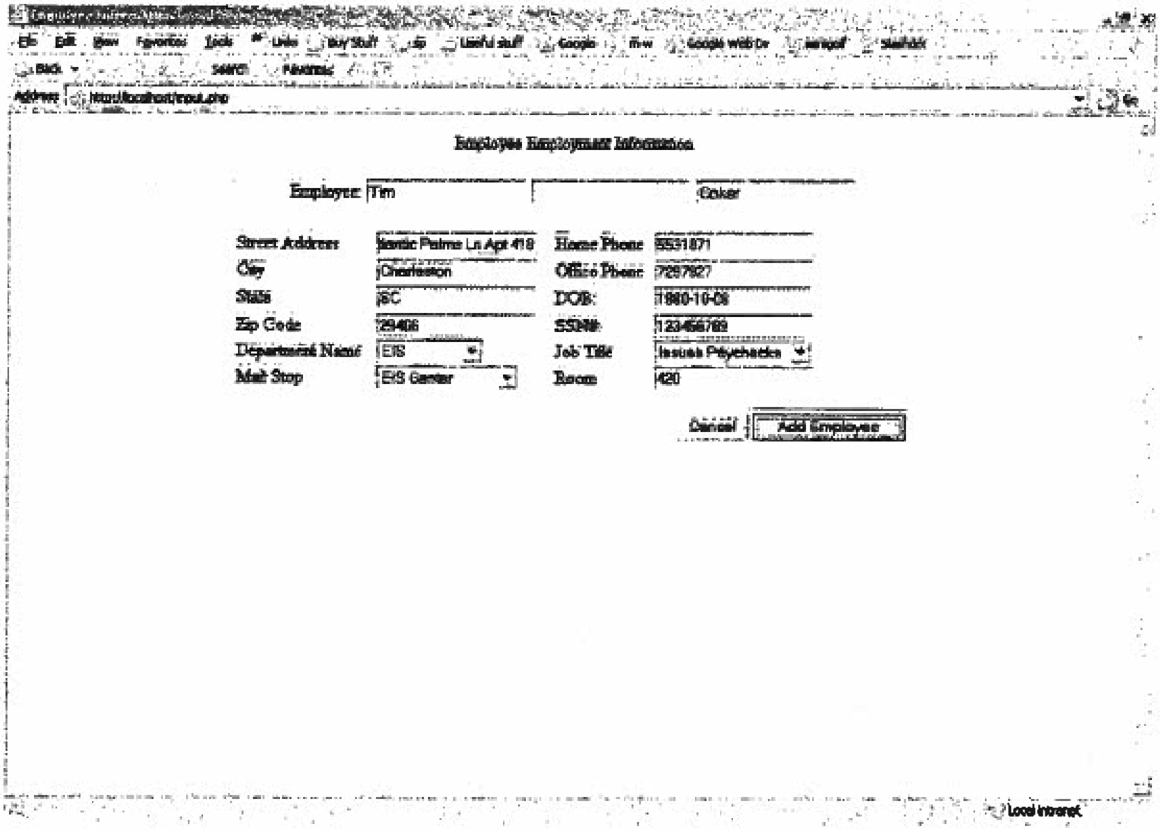



**Employee Self Service System**

*all searches are by last name


user
Web Browser
(Internet Explorer)
HTML
Web Server
(Apache)
PHP
Text
PHP Processor
SQL
MySql


Employee Self-Service System
Add New Employee
Local intranet


Employee Employment Information
Employee: Tim
Coker
Street Address
Home Phone
5531871
City
Charleston
Office Phone
7297927
State
SC
DOB:
Zip Code
29406
SSN#:
123456789
Department Name
EIS
Job Title
Mail Stop
EIS Center
Room
420
Cancel
Add Employee
Local intranet

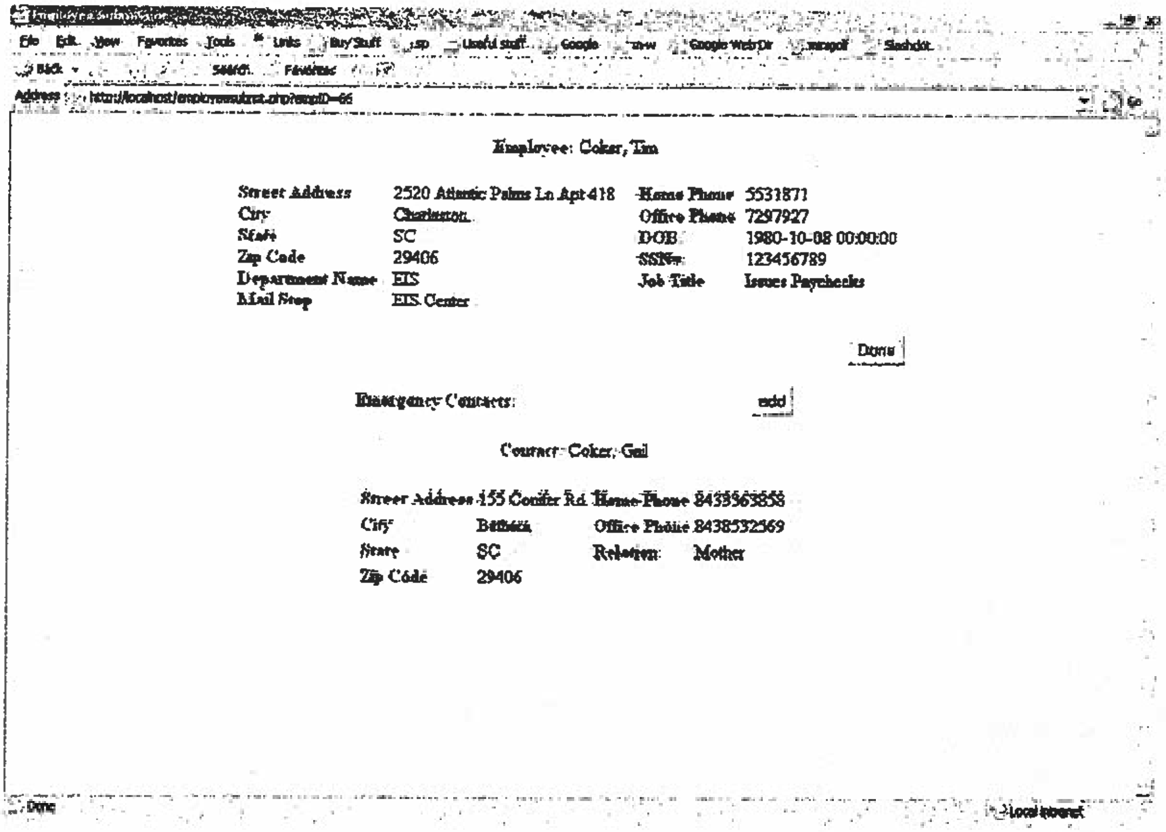
Emergency Contact Information

## Emergency Contact Information

Contact Name: Gail | | Coker

| | | | |
|---|---|---|---|
| Street Address | 155 Conifer Rd. | Evening Phone | 8433383858 |
| City | Bethera | Daytime Phone | 8438532589 |
| State | SC | Relation: | Mother |
| Zip Code | 29406 | | |

Cancel | Submit



File Edit View Favorites Tools Links Buy Stuff Useful stuff Google Google Web Dir Slashdot

Back Search Favorites

Address http://localhost/employeesubmit.php?empID=66

Employee: Coker, Tim

| | | | |
|---|---|---|---|
| **Street Address** | 2520 Atlantic Palms Ln Apt 418 | **Home Phone** | 5531871 |
| **City** | Charleston | **Office Phone** | 7297927 |
| **State** | SC | **DOB** | 1980-10-08 00:00:00 |
| **Zip Code** | 29406 | **SSN#** | 123456789 |
| **Department Name** | EIS | **Job Title** | Issues Paychecks |
| **Mail Stop** | EIS Center | | |

Done

**Emergency Contacts:** add

Contact: Coker, Gail

| | | | |
|---|---|---|---|
| **Street Address** | 155 Conifer Rd. | **Home Phone** | 8435563858 |
| **City** | Bethera | **Office Phone** | 8438532569 |
| **State** | SC | **Relation:** | Mother |
| **Zip Code** | 29406 | | |

Done Local intranet

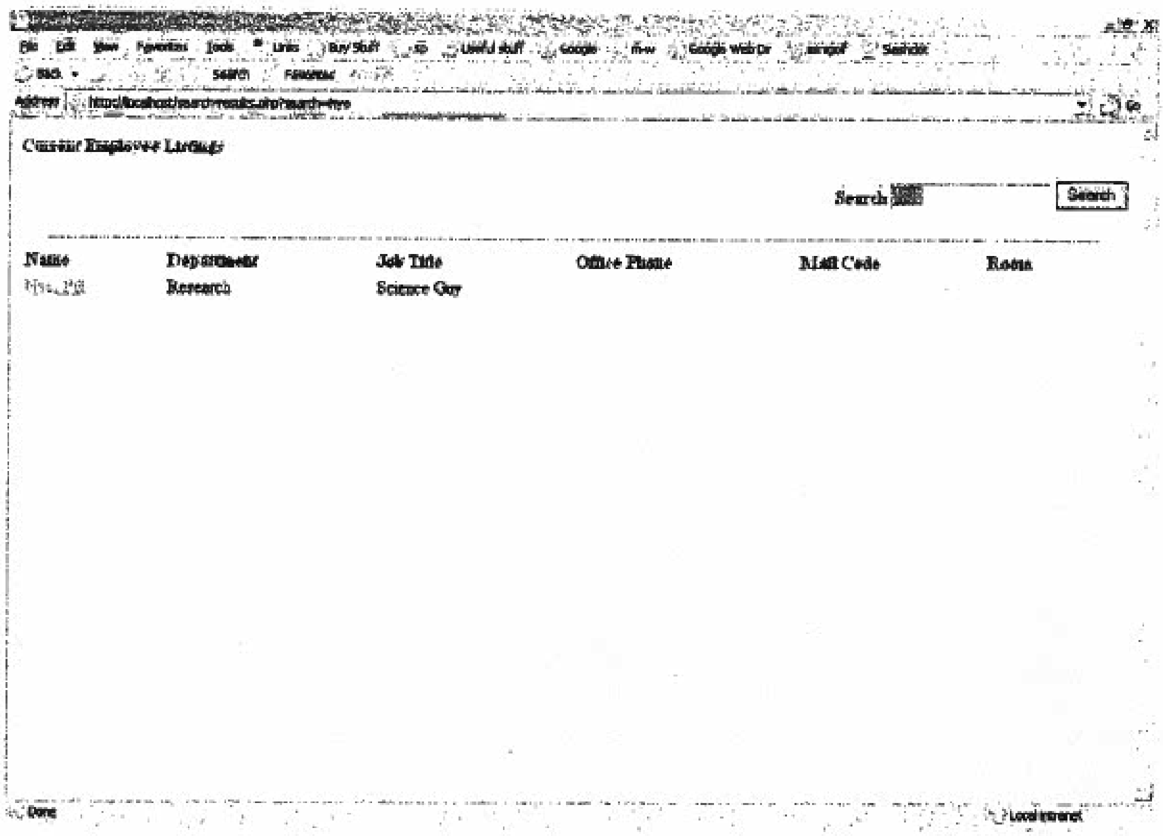
Address http://localhost/searchresults.php

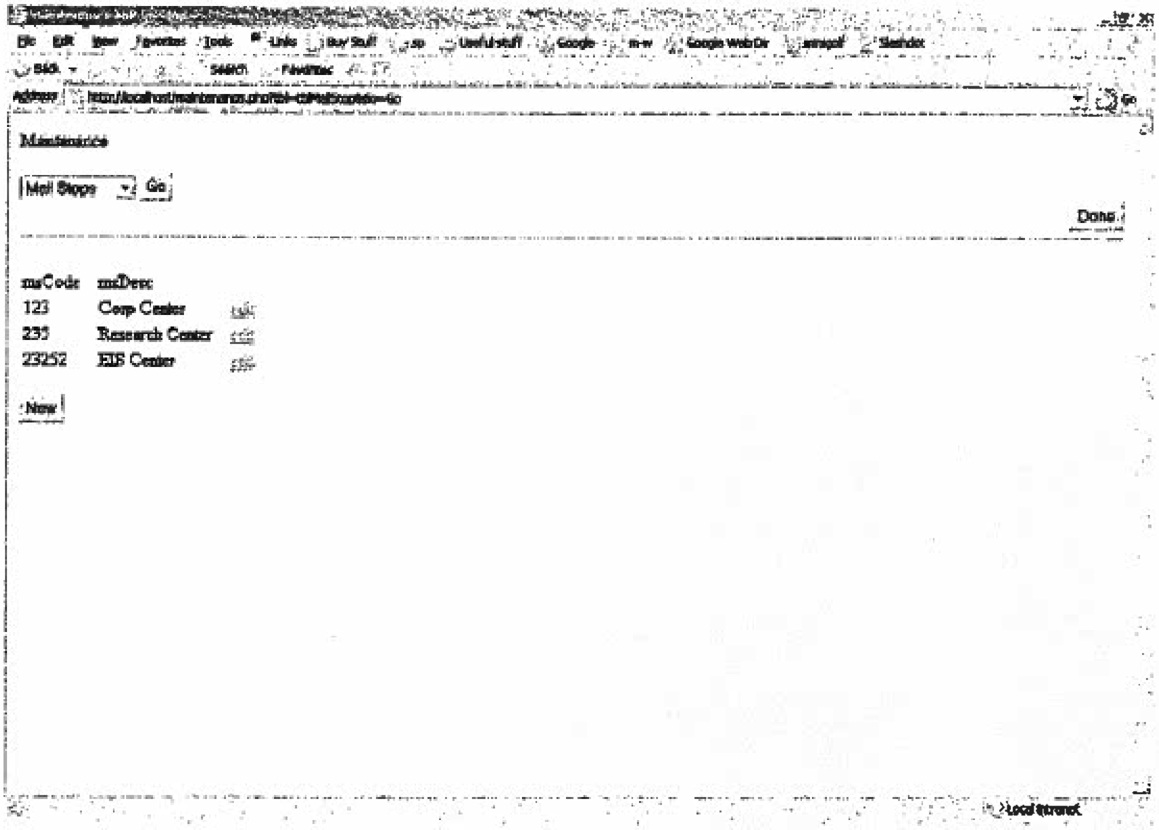
**Current Employee Listings**

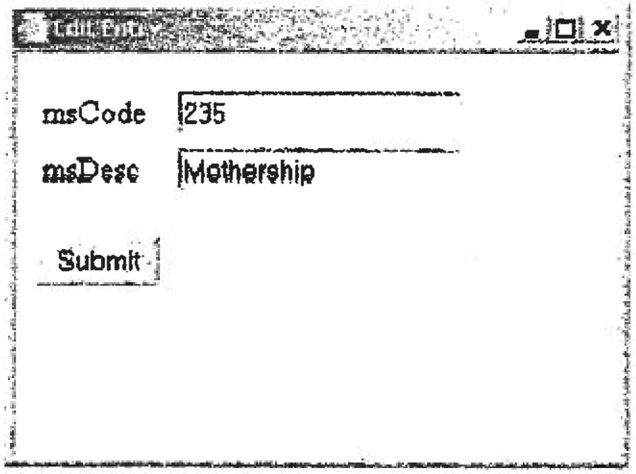
Search [ ] Search

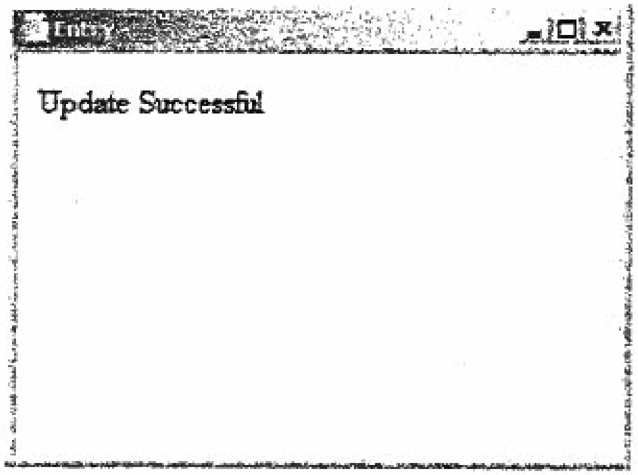
| Name | Department | Job Title | Office Phone | Mail Code | Room |
|---|---|---|---|---|---|
| [illegible] | Retardation | asdf | 1-843-740-2002 | | |
| [illegible] | Marketing | asdf | 123-456-7890 | Corp Center | |
| [illegible] | Research | Science Guy | | | |
| Last, First | zxcv | asdf | | Corp Center | |
| [illegible] | EIS | Pointy Haired Boss | | Corp Center | |
| [illegible] | EIS | Pointy Haired Boss | 234-5543 | Corp Center | |
| [illegible] | EIS | Issues Paychecks | 729-7927 | EIS Center | 420 |

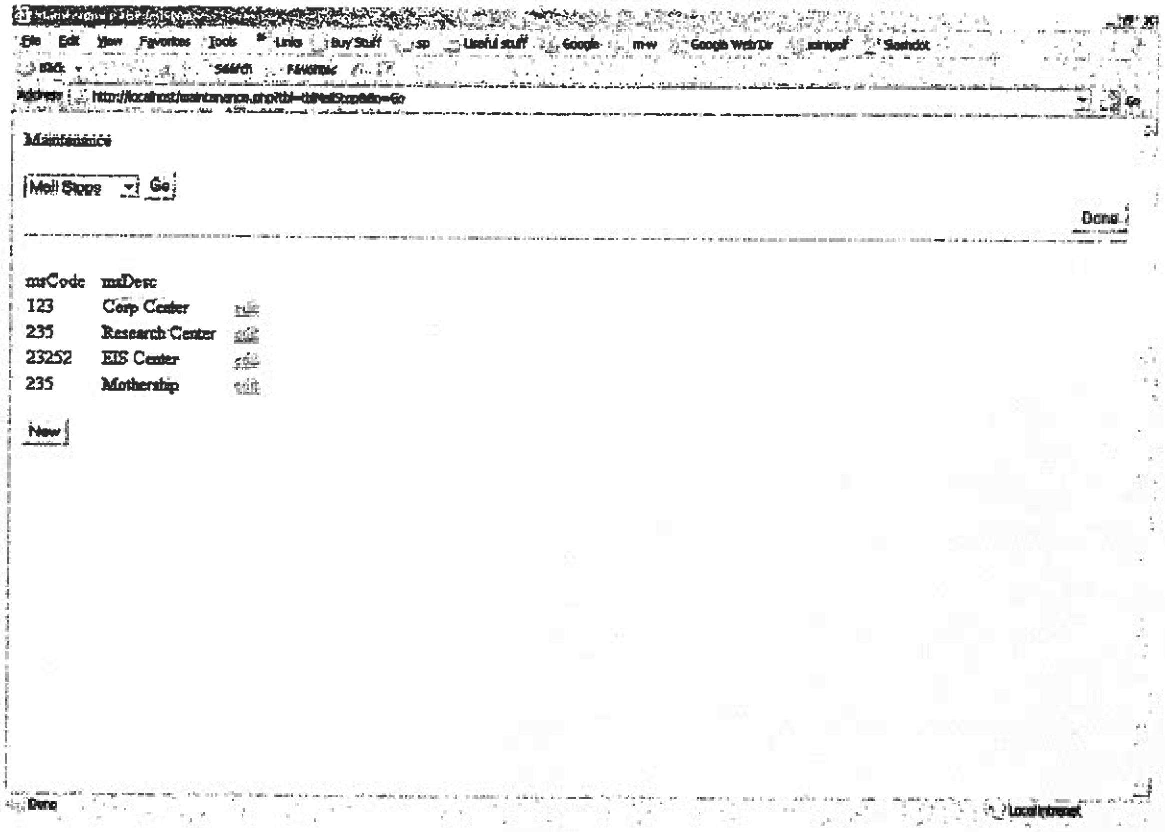
Local intranet

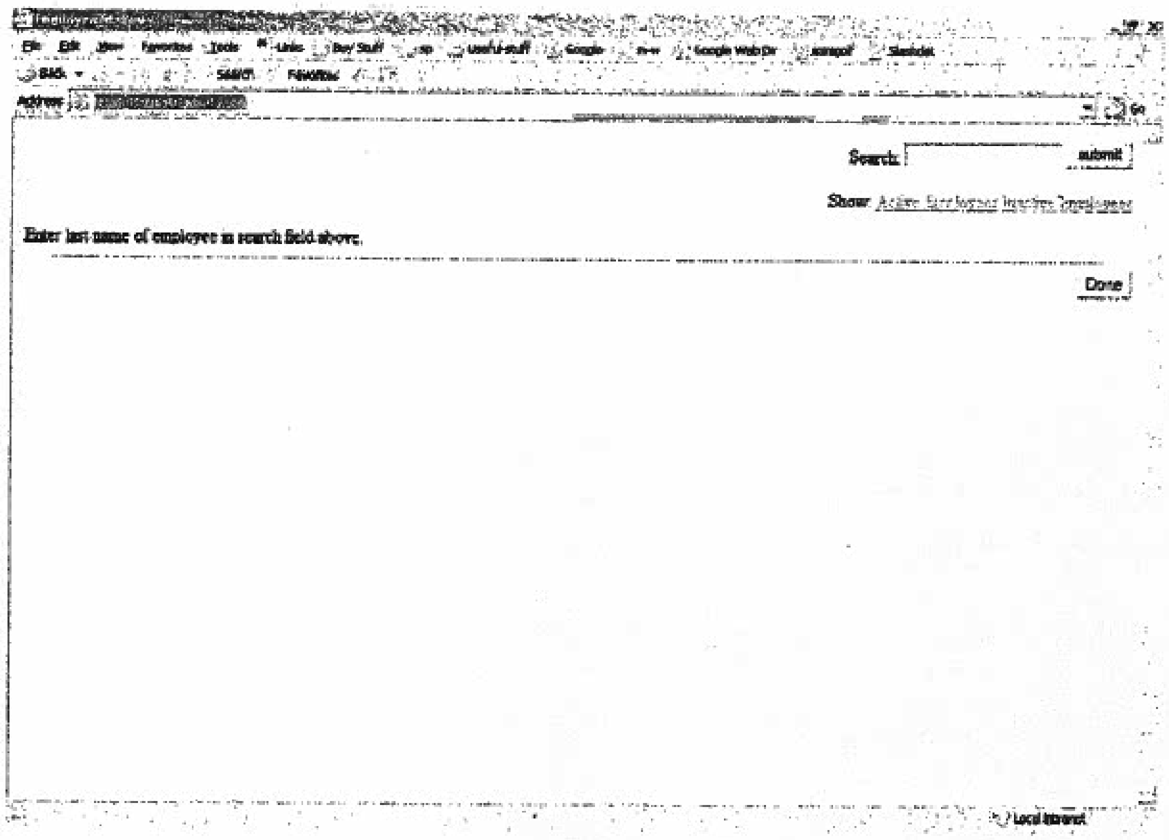

File Edit View Favorites Tools Links Buy Stuff Useful stuff Google Google Web Dir
Back Search Favorites
Address http://localhost/search-results.php?search=
Go
Current Employee Listings
Search
Search
Name
Department
Job Title
Office Phone
Mail Code
Room
Research
Science Guy
Done
Local intranet

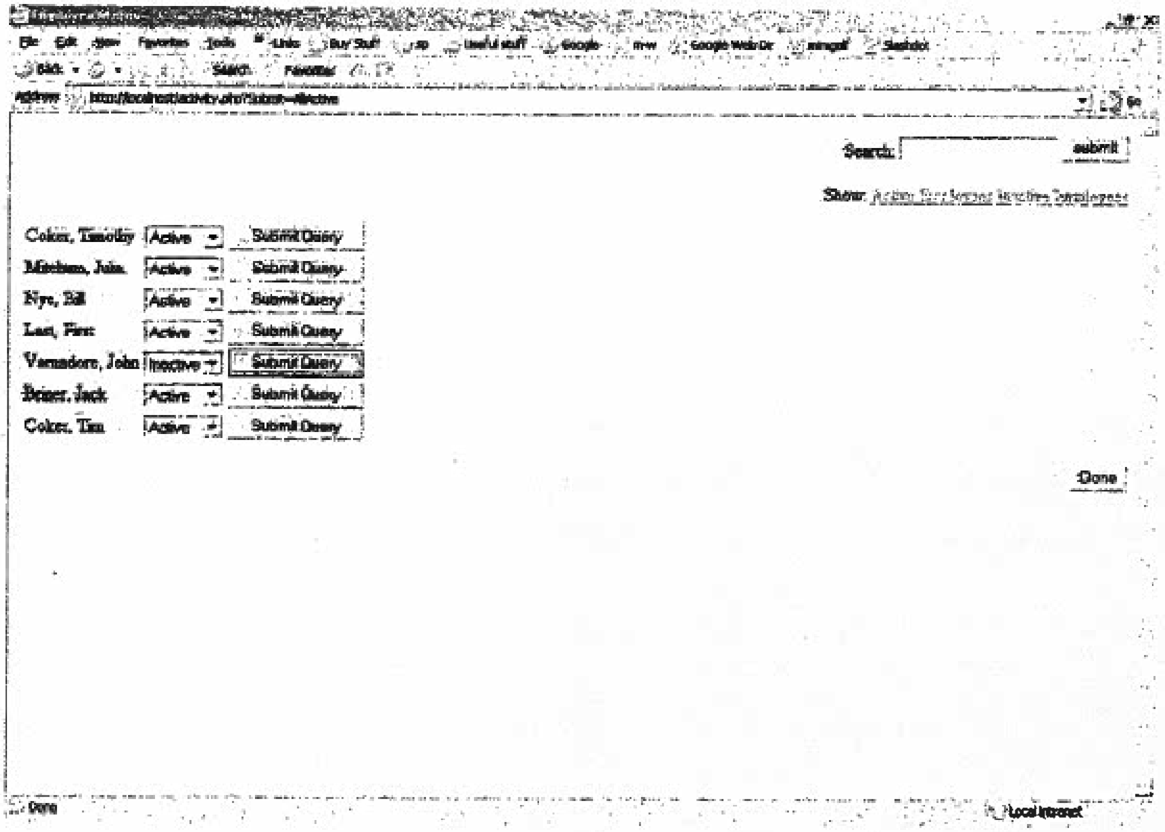

Maintenance
Mail Stops
Go
Done
msCode msDesc
123 Corp Center edit
235 Research Center edit
23252 EIS Center edit
New
Local intranet


Edit Entry
msCode
235
msDesc
Mothership
Submit

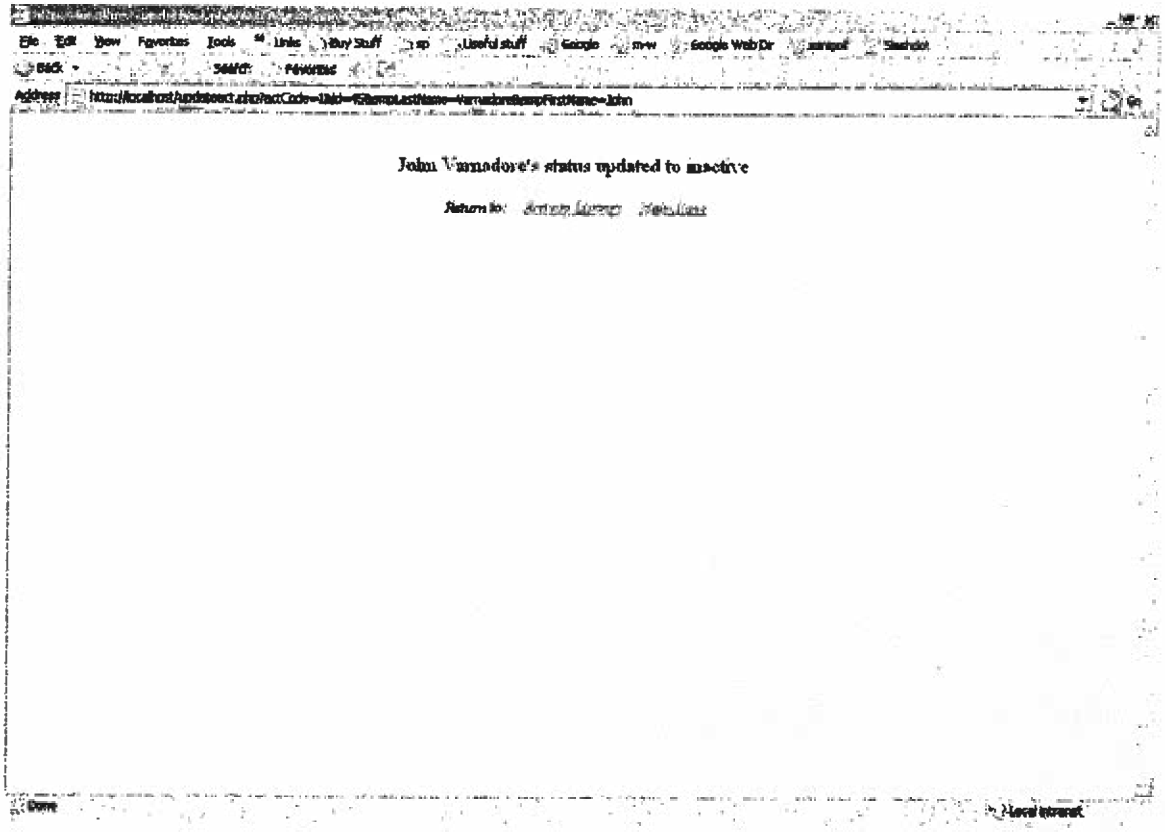

Entry
Update Successful


File Edit View Favorites Tools Links Buy Stuff Useful stuff Google Google Web Dir Slashdot
Address http://localhost/maintenance.php
Maintenance
Mall Stops Go
Done
msCode msDesc
123 Corp Center edit
235 Research Center edit
23252 EIS Center edit
235 Mothership edit
New
Done
Local intranet


File Edit View Favorites Tools
Links
Useful-stuff
Google
Slashdot
Search Favorites
Address
Go
Search:
submit
Show: Active Employees Inactive Employees
Enter last name of employee in search field above.
Done
Local intranet



File Edit View Favorites Tools
Search:
submit
Show:
Coker, Timothy
Active
Submit Query
Mitcham, Julia
Active
Submit Query
Nye, Bill
Active
Submit Query
Last, First
Active
Submit Query
Varnadore, John
Inactive
Submit Query
Brizer, Jack
Active
Submit Query
Coker, Tim
Active
Submit Query
Done
Done
Local intranet


File Edit View Favorites Tools Links Buy Stuff Useful stuff Google Google Web Dir Slashdot
Back Search Favorites
Address
John Vannadore's status updated to inactive
Return to:
Done
Local intranet

```
<html>
<? include("include/include.php"); ?>

<head>
<title>Employee Status</title>
</head>

<body>
<div align="right">
   <form>Search: <input name=search value=<?echo$search?>>
   <input type=submit name=Submit value=submit></form>
   Show:
   <a href="activity.php?Submit=AllActive">Active Employees</a>
   <a href="activity.php?Submit=AllInActive">Inactive Employees</a>

</div>
<div align="center">
   <hr width="95%" size="0">
</div>

<table>
<?
      /* *
         *
         * we have a name specified to search for
         *
         */
      if (isset($search) && $search!="") {
         $tmpquery = "SELECT * FROM tblemp WHERE empLastName =
         \"".$search."\"";
         $tmpresult = mysql_query($tmpquery) or die("Query failed
         when getting emps");
         while ($line = mysql_fetch_assoc($tmpresult)) { ?>
            <form method="get" action="updateact.php">
               <tr><td>
                  <? echo $line["empLastName"].",
                  ".$line["empFirstName"]."
                  ".$line["empMiddleName"]; ?>
                  <td>
                     <select name="actCode" >
                        <? if ($line["empActCode"]==0) {?>
                           <option selected value=0>Active</option>
                           <option value=1>Inactive</option>
                        <? } else { ?>
                           <option value=0>Active</option>
                           <option selected
                           value=1>Inactive</option>
                        <? } //end if ?>
                        <input name="id" type=hidden value="<?echo
                        $line["empID"];?>">
                        <input type=submit><br>
                     </select>
                     <input type=hidden
                     value="<?echo$line["empLastName"]?>"
                     name=empLastName>
                     <input type=hidden
                     value="<?echo$line["empFirstName"]?>"
                     name=empFirstName>
                  </td>
               </tr>
            </form>
         <? } //end while

      /* *
         *
         * we are showing all active employees
         *
         */

      } else if ($Submit=="AllActive") {
         $tmpquery = "SELECT * FROM tblemp WHERE empActCode =
```

```
<html>
<? include("include/include.php"); ?>
<head>
<title>Emergency Contact Information</title>
</head>

<body>
   <? if (isset($Submit)) {
      $query="INSERT INTO tblEmgCntct ";
      $query=$query."(cntctFirstName, cntctMiddleName,
      cntctLastName, cntctStreetAddress, cntctCity, cntctState,
      cntctZip, cntctHomePhone, cntctWorkPhone, cntctRelation,
      cntctEmpID) ";
      $query=$query." VALUES ";
      $query=$query."('$empFirstName', '$empMiddleName',
      '$empLastName', '$empStreetAddress', '$empCity', '$empState',
      '$empZip', '$empHomePhone', '$empWorkPhone', '$empRelation',
      '$empID') ";
      $result=mysql_query($query) or die("error on inserting new
      contact");
      ?>
      <font size=+1>Emergency Contact added Successfully</font>
      <?
         echo"<script>";

         echo"window.opener.location=\"employeesubmit.php?empID=".$e
         mpID."\";";

         echo"window.setTimeout(\"window.close();window.opener.focus
         ();\",750);";
         echo"</script> ";
      ?>
      <input type="hidden" value="Not Submitted" name="Submit">
   <? } else { ?>
      <div align="center">Emergency Contact Information</div>
      <form name="info" method="get">
         <div align="center">
         <table>
            <tr>
               <td>Contact Name:</td>
               <td>
                  <input type="text" name="empFirstName"
                  value="First" onFocus="if (this.value=='First')
                  this.value='';">
                  <input type="text" name="empMiddleName"
                  value="Middle" onFocus="if (this.value=='Middle')
                  this.value='';">
                  <input type="text" name="empLastName" value="Last"
                  onFocus="if (this.value=='Last') this.value='';">
               </td>
            </tr>
         </table>
        </div>
        <br>
         <table width="600" border="0" cellspacing="0"
         cellpadding="0" height="0%" align="center">
            <tr>
               <td>Street Address</td>
               <td>
                  <input type="text" name="empStreetAddress">
               </td>
               <td>Evening Phone</td>
               <td>
                  <input type="text" name="empHomePhone">
               </td>
            </tr>
            <tr>
               <td>City</td>
               <td>
                  <input type="text" name="empCity">
               </td>
```

```
                <td>Daytime Phone</td>
                <td>
                    <input type="text" name="empWorkPhone">
                </td>
            </tr>
            <tr>
                <td>State</td>
                <td>
                    <input type="text" name="empState">
                </td>
                <td>Relation:</td>
                <td>
                    <input type="text" name="empRelation">
                </td>
            </tr>
            <tr>
                <td>Zip Code</td>
                <td>
                    <input type="text" name="empZip">
                </td>
            </tr>
            <tr><td> </td></tr>
            <tr>
                <td></td>
                <td></td>
                <td></td>
                <td>
                    <div align="right">
                        <input type="button" name="Cancel"
                        value="Cancel" onClick="window.close()">
                        <input type="Submit" name="Submit"
                        value="Submit">
                    </div>
                </td>
            </tr>
        </table>
        <input type="hidden" name="empID" value="<?echo$empID?>">
    </form>
  <? } //end if block?>
</body>
</html>
```

```
<html>
<? include("include/include.php"); ?>
<head>
<title>Employee Submission</title>
</head>

<body>
   <? if ($Submit == "Add Employee") { //update employee
      $query="INSERT INTO tblEmp ";
      $query=$query."(empFirstName, empMiddleName, empLastName,
      empStreetAddress, empCity, empState, empZip, empHomePhone,
      empWorkPhone, empDOB, empMailStopID, empDeptID, empJobID,
      empSSN, empRoom) ";
      $query=$query." VALUES ";
      $query=$query."('$empFirstName', '$empMiddleName',
      '$empLastName', '$empStreetAddress', '$empCity', '$empState',
      '$empZip', '$empHomePhone', '$empWorkPhone', '$empDOB',
      '$empMailStopID', '$empDeptID', '$empJobID', '$empSSN',
      '$empRoom') ";
      $result=mysql_query($query) or die("error...");
   }
   ?>
   <? //get record just updated and prompt for emergency contacts
      $query="SELECT * from tblEmp where
      (empFirstName=\"$empFirstName\" AND
      empMiddleName=\"$empMiddleName\" AND
      empLastName=\"$empLastName\" AND
      empStreetAddress=\"$empStreetAddress\" AND
      empCity=\"$empCity\" AND empState=\"$empState\" AND
      empZip=\"$empZip\" AND empHomePhone=\"$empHomePhone\" AND
      empDOB=\"$empDOB\" AND empMailStopID=\"$empMailStopID\" AND
      empDeptID=\"$empDeptID\" AND empJobID=\"$empJobID\" AND
      empSSN=\"$empSSN\") OR (empID=\"$empID\")";
      $result=mysql_query($query) or die("error on getting
      submitted record");
      while ($line = mysql_fetch_assoc($result)) { ?>
         <?$empID=$line["empID"];?>
         <div align="center">
            <table>
               <tr>
                  <td><b>Employee:</b></td>
                  <td>
                     <? echo $line["empLastName"].",
                     ".$line["empFirstName"]."
                     ".$line["empMiddleName"]; ?>
                  </td>
               </tr>
            </table>
         </div>
        <br>
         <table width="600" border="0" cellspacing="0"
         cellpadding="0" height="0%" align="center">
            <tr>
               <td><b>Street Address</b></td>
               <td>
                  <? echo $line["empStreetAddress"]; ?>
               </td>
               <td><b>Home Phone</b></td>
               <td>
                  <? echo $line["empHomePhone"]; ?>
               </td>
            </tr>
            <tr>
               <td><b>City</b></td>
               <td>
                  <? echo $line["empCity"]; ?>
               </td>
               <td><b>Office Phone</b></td>
               <td>
                  <? echo $line["empWorkPhone"]; ?>
               </td>
```

```
            </tr>
            <tr>
                <td><b>State</b></td>
                <td>
                    <? echo $line["empState"]; ?>
                </td>
                <td><b>DOB:</b></td>
                <td>
                    <? echo $line["empDOB"]; ?>
                </td>
            </tr>
            <tr>
                <td><b>Zip Code</b></td>
                <td>
                    <? echo $line["empZip"]; ?>
                </td>
                <td><b>SSN#:</b></td>
                <td>
                    <? echo $line["empSSN"]; ?>
                </td>
            </tr>
            <tr>
                <td><b>Department Name</b></td>
                <td>
                    <? //get deptdesc from tbldept using deptid stored in tblemp
                        $tmpquery = "SELECT * FROM tblDept WHERE deptID = ".$line['empDeptID'];
                        $tmpresult = mysql_query($tmpquery) or die("Query failed when getting deptdesc");
                        while ($tmpline = mysql_fetch_assoc($tmpresult))
                            echo $tmpline[deptDesc] ;
                    ?>
                </td>
                <td><b>Job Title</b></td>
                <td>
                    <?
                        $tmpquery = "SELECT * FROM tblJobDesc WHERE jobID=".$line['empJobID'];
                        $tmpresult = mysql_query($tmpquery) or die("Query failed when getting jobdesc");
                        while ($tmpline = mysql_fetch_assoc($tmpresult))
                            echo $tmpline[jobDesc];
                    ?>
                </td>
            </tr>
            <tr>
                <td><b>Mail Stop</b></td>
                <td>
                    <?
                        $tmpquery = "SELECT * FROM tblMailStop WHERE msID = ".$line["empMailStopID"];
                        $tmpresult = mysql_query($tmpquery) or die("Query failed when getting mailstop");
                        while ($tmpline = mysql_fetch_assoc($tmpresult))
                            echo $tmpline[msDesc];
                    ?>
                </td>
                <td> </td>
            </tr>
        </table>
    <?}?>
<br>
<div align="center"><table width="60%"><tr><td><div align="right"><input type=button value=Done onclick="window.location='main.htm'"></div></td></tr></table></div>
<hr width="65%" size=1>
<div align=center>
```

```
<table width=40%><tr><td>
<font><b>Emergency Contacts:</font>
<td align=right><input type=button name="submit" value="add"
onclick="window.open('emgcontact_input.php?empID=<?echo$empID?>',''
,'menubar=no,toolbar=no,resizable,location=no,status=no,width=700,h
eight=250')">
</table>
</div>      <? //get list of emergency contacts
        $query="SELECT * from tblEmgCntct where
        cntctEmpID=\"$empID\"";
        $result=mysql_query($query) or die("error on getting
        emergency contact records");
        while ($line = mysql_fetch_assoc($result)) { ?>
           <div align="center">
           <hr width="45%" size=1>
              <table>
                 <tr>
                    <td><b>Contact:</b></td>
                    <td>
                       <? echo $line["cntctLastName"].",
                       ".$line["cntctFirstName"]."
                       ".$line["cntctMiddleName"]; ?>
                    </td>
                 </tr>
              </table>
           </div>
          <br>
           <table width="" border="0" cellspacing="2" cellpadding="1"
           height="0%" align="center">
              <tr>
                 <td><b>Street Address</b></td>
                 <td>
                    <? echo $line["cntctStreetAddress"]; ?>
                 </td>
                 <td><b>Home Phone</b></td>
                 <td>
                    <? echo $line["cntctHomePhone"]; ?>
                 </td>
              </tr>
              <tr>
                 <td><b>City</b></td>
                 <td>
                    <? echo $line["cntctCity"]; ?>
                 </td>
                 <td><b>Office Phone</b></td>
                 <td>
                    <? echo $line["cntctWorkPhone"]; ?>
                 </td>
              </tr>
              <tr>
                 <td><b>State</b></td>
                 <td>
                    <? echo $line["cntctState"]; ?>
                 </td>
                 <td><b>Relation:</b></td>
                 <td>
                    <? echo $line["cntctRelation"]; ?>
                 </td>
              </tr>
              <tr>
                 <td><b>Zip Code</b></td>
                 <td>
                    <? echo $line["cntctZip"]; ?>
                 </td>
              </tr>
           </table>
        <?}?>
</body>
</html>
```

```
<html>
<? include("include/include.php"); ?>
<head>
<title>Employee Information</title>
</head>

<body>
<div align="center">Employee Employment Information</div>
<form name="info" method="get" action="employeesubmit.php">
   <div align="center">
   <table>
      <tr>
         <td>Employee:</td>
         <td>
            <input type="text" name="empFirstName" value="First" onFocus="if (this.value=='First') this.value='';">
            <input type="text" name="empMiddleName" value="Middle" onFocus="if (this.value=='Middle') this.value='';">
            <input type="text" name="empLastName" value="Last" onFocus="if (this.value=='Last') this.value='';">
         </td>
      </tr>
   </table>
  </div>
  <br>
   <table width="600" border="0" cellspacing="0" cellpadding="0" height="0%" align="center">
      <tr>
         <td>Street Address</td>
         <td>
            <input type="text" name="empStreetAddress">
         </td>
         <td>Home Phone</td>
         <td>
            <input type="text" name="empHomePhone">
         </td>
      </tr>
      <tr>
         <td>City</td>
         <td>
            <input type="text" name="empCity">
         </td>
         <td>Office Phone</td>
         <td>
            <input type="text" name="empWorkPhone">
         </td>
      </tr>
      <tr>
         <td>State</td>
         <td>
            <input type="text" name="empState">
         </td>
         <td>DOB:</td>
         <td>
            <input type="text" name="empDOB" value="yyyy-mm-dd" onfocus="this.value=''">
         </td>
      </tr>
      <tr>
         <td>Zip Code</td>
         <td>
            <input type="text" name="empZip">
         </td>
         <td>SSN#:</td>
         <td>
            <input type="text" name="empSSN">
         </td>
      </tr>
      <tr>
         <td>Department Name</td>
         <td>
```

```
<?
	//establish connection
	$link = mysql_connect($HOST, $USER, $PASS)
	 or die("Could not connect");
	mysql_select_db("esss")
	 or die("Could not select database");

	function parsePhone($b) {
	// parses 11 or 10 digit phonenumbers w/ hyphens
		$c = "";
		for ($i=0; $i < strlen($b); $i++) {
			if ((strlen($b) == 11) && (($i == 0) || ($i == 3) || ($i == 6))) {
				$c .= $b{$i}."-";
			} else if ((strlen($b) == 10) && (($i == 2) || ($i == 5))) {
				$c .= $b{$i}."-";
			} else if ((strlen($b) == 7) && ($i == 2)) {
				$c .= $b{$i}."-";
			} else {
				$c .= $b{$i};
			}
		}
		return $c;
	}

	function getData($rtfld, $cmpfld, $tbl, $id, $txt) {
	//gets description for related ID, ie, job description for job id, etc....
		$tmpquery = "SELECT * FROM $tbl WHERE $cmpfld = $id";
		$tmpresult = mysql_query($tmpquery)
		 or die("Query failed when getting ". $txt);
		$tmpline = mysql_fetch_assoc($tmpresult);
		return ($tmpline[$rtfld]);
	}

	$maintTbls[] = "tblSite";		$maintTbls_desc[] = "Sites";
	$maintTbls[] = "tblRoom";		$maintTbls_desc[] = "Rooms";
	$maintTbls[] = "tblMailStop";	$maintTbls_desc[] = "Mail Stops";
	$maintTbls[] = "tblJobDesc";	$maintTbls_desc[] = "Job Titles";
	$maintTbls[] = "tblDept";		$maintTbls_desc[] = "Departments";
	$maintTbls[] = "tblBldg";		$maintTbls_desc[] = "Buildings";
?>
```

```
            <select name="empDeptID">
               <?
                  $tmpquery = "SELECT * FROM tblDept";
                  $tmpresult = mysql_query($tmpquery) or die("Query
                  failed when getting deptdesc");
                  echo $tmpresult;
                  while ($line = mysql_fetch_assoc($tmpresult))
                     echo "<option value=\"$line[deptID]\"
                     >$line[deptDesc]</option>\n\t";
               ?>
            </select>
         </td>
         <td>Job Title</td>
         <td>
            <select name="empJobID">
               <?
                  $tmpquery = "SELECT * FROM tblJobDesc";
                  $tmpresult = mysql_query($tmpquery) or die("Query
                  failed when getting jobdesc");
                  while ($line = mysql_fetch_assoc($tmpresult))
                     echo "<option value=\"$line[jobID]\"
                     >$line[jobDesc]</option>\n\t";
               ?>
            </select>
         </td>
      </tr>
      <tr>
         <td>Mail Stop</td>
         <td>
            <select name="empMailStopID">
               <?
                  $tmpquery = "SELECT * FROM tblMailStop";
                  $tmpresult = mysql_query($tmpquery) or die("Query
                  failed when getting mailstop");
                  while ($line = mysql_fetch_assoc($tmpresult))
                     echo "<option value=\"$line[msID]\"
                     >$line[msDesc]</option>\n\t";
               ?>
            </select>
         </td>
         <td>Room</td>
         <td>
            <input type="text" name="empRoom">
         </td>
      <td> </td></tr>
      <tr><td> </td></tr>
      <tr>
         <td></td>
         <td></td>
         <td></td>
         <td>
            <div align="right">
               <input type="button" name="Cancel" value="Cancel"
               onClick="history.back()">
               <input type="submit" name="Submit" value="Add
               Employee">
            </div>
         </td>
      </tr>
   </table>
</form>
</body>
</html>
```

```
<html>
<head>
<title>ESSS</title>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">
</head>

<body bgcolor="#FFFFFF" text="#000000">
<p align="center"><font size="+3">Employee Self-Service System</font></p>
<p align="center"> </p>
<p align="center"><a href="input.php">Add New Employee</a></p>
<p align="center"><a href="search-results.php">View Listings</a></p>
<p align="center"><a href="maintenance.php">Table Maintenance</a></p>
<p align="center"><a href="activity.php">Employee Status</a></p>
</body>
</html>
```

```
<html>
<? include("include/include.php"); ?>
<head>
<title>Maintenance <?echo $tbl?></title>
<meta http-equiv="Content-Type" content="text/html;
charset=iso-8859-1">
<script>
   function edit(tbl,rid,fld) {

      window.open("mini_maint.php?tbl="+tbl+"&id="+rid+"&fld="+fld+"
      &update=Edit","","menubar=no,toolbar=no,resizable,location=no,
      status=no,width=300,height=200");
   }
   function add(tbl,rid,fld) {

      window.open("mini_maint.php?tbl="+tbl+"&id="+rid+"&fld="+fld+"
      &new=New","","menubar=no,toolbar=no,resizable,location=no,stat
      us=no,width=300,height=200");
   }
</script>
</head>

<body bgcolor="#FFFFFF" text="#000000">
<p>Maintenance</p>
<form name="tblSelect" method="get" action="">
   <select name="tbl">
   <?
      for ($i = 0; $i < count($maintTbls); $i++) {
         echo "<option ";
            if($tbl==$maintTbls[$i]) {
               echo "selected";
            }
            echo "
            value=\"$maintTbls[$i]\">$maintTbls_desc[$i]</option>\n
                  ";
      }
   ?>
   </select>
   <input type="submit" name="Go" value="Go">
<div align=right><input type=button value=Done
onclick="window.location='main.htm'"></div>
<hr width="100%">
</form>
<?
   if (isset($Go)) {
      $query = "SELECT * FROM $tbl";
      $result = mysql_query($query) or die("Query failed");
      echo "<table>\n   "; //start of form and table in result
         $line = mysql_fetch_row($result);  //getting table headings
         echo "<tr>\n      ";
         for ($i=1; $i<count($line); $i++) { //start at 1 to skip id
         field (internal data)
            echo "<td
            class=\"heading\">".mysql_field_name($result,$i)." 
              </td>\n       ";
         }
         mysql_data_seek($result,0); //reset to 0
         while ($line = mysql_fetch_row($result)) { //output column
         information
            echo "</tr><tr>\n       ";
            for ($i=1; $i<count($line); $i++) { //start at 1 to skip
            id field (internal data)
               echo "<td>$line[$i]   </td>\n       ";
            }
            echo "<td><a class=\"edit\"
            href=\"javascript:edit('$tbl',$line[0],'".mysql_field_na
            me($result,0)."');\">edit</a></td>\n";
         }
      echo "</table>\n";
      mysql_data_seek($result,0); //reset to 0
      $line=mysql_fetch_row($result);
```

```
        echo "<br><input type=\"button\" value=\"New\"
        onClick=\"add('$tbl','$line[0]','".mysql_field_name($result,0)
        ."');\"></form>\n";
    }
?>
</body>
</html>
```

```
<html>
<? include("include/include.php"); ?>
<?
   //this block retrieves the first record of the table in
   //question when we load the page
   $query="SELECT * FROM $tbl WHERE $fld = $id";
   $result=mysql_query($query) or die("Error");
   $line=mysql_fetch_row($result);

   function printQuerystringinfo($tbl,$id,$fld) {
      //this spits out the fields passed in the query string
      //we need to keep this for the update statement when this page is
      //submitted back to itself (the if isset($add), isset($Submit) blocks
      //there may be a better way of doing this...
      echo "<input type=\"hidden\" value=\"$tbl\" name=\"tbl\">\n";
      echo "<input type=\"hidden\" value=\"$fld\" name=\"fld\">\n";
      echo "<input type=\"hidden\" value=\"$id\" name=\"id\">\n";
   }

   function printInputFields($line,$result,$test) {
      //this prints the actual fields on the form
      //its used when we are entering a new record or updating an old one,
      //if we're updating an old one, we want the values in the fields, hence
      //the if block in the middle of outputting the input tag
      echo "<form method=\"get\">\n<table>";
      for ($i=1; $i<count($line);$i++) { //start at 1 to skip id field (internal data)
         echo "<tr>\n";
         echo "<td>".mysql_field_name($result,$i)."   </td>\n";
         echo "<td><input type=\"text\" ";
         if ($test == "Edit") {
            echo "value=\"$line[$i]\" ";
         }
         echo "name=\"vals[]\"></td></tr>\n";
      }
      echo "</table>\n<p><input type=submit value=\"Submit\" name=\"Submit\">\n";
   }
?>
<title><?echo $update,$new?> Entry</title>
<script>
   function exit() {
      //javascript statements that refreshes the opening page and closes the current window
      //and sets the focus back to the opening page used when this form is submitted back
      //to itself and the databases are submitted
      window.opener.location=window.opener.location;
      window.setTimeout("window.close();window.opener.focus();",750);
   }
</script>
<body>
<?
   if (isset($update)) {
      //updating record - we have a record we want to update
      printInputFields($line,$result,$update);
      printQuerystringInfo($tbl,$id,$fld);
   } else if (isset($new)) {
      //adding new record - we are adding a new record to the table
      printInputFields($line,$result,$update);
      printQuerystringInfo($tbl,$id,$fld);
      echo "<input type=\"hidden\" value=\"add\" name=\"add\">\n"; //set variable to keep track of state
```

```
    } else if (isset($add)) {
      //the new record has been submitted, so we insert into the table.  first
      //we build the query then submit it and echo Update Successful and exit.
      $query="INSERT INTO $tbl (";
      for ($i=0; $i<count($vals); $i++) {
        if ($i == count($vals)-1) {
          $query=$query."`".mysql_field_name($result,$i+1)."`) ";
        } else {
          $query=$query."`".mysql_field_name($result,$i+1)."`, ";
        }
      }
      $query=$query." VALUES (";
      for ($i=0; $i<count($vals); $i++) {
        if ($i == count($vals)-1) {
          $query=$query."'".$vals[$i]."') ";
        } else {
          $query=$query."'".$vals[$i]."', ";
        }
      }
      $result=mysql_query($query) or die("error...");
      echo "Addition Successful\n";
      echo "<script>exit();</script>";
    } else if (isset($Submit)) {
      //the record has been submitted for update, so we update the existing one in the table.
      //first we build the query then submit it and echo Update Successful and exit.
      $query="UPDATE $tbl SET ";
      for ($i=0; $i<count($vals); $i++) {
        if ($i == count($vals)-1) {
          $query=$query."`".mysql_field_name($result,$i+1)."`='".$vals[$i]."' ";
        } else {
          $query=$query."`".mysql_field_name($result,$i+1)."`='".$vals[$i]."', ";
        }
      }
      $query=$query." WHERE `$fld` = '$id'";
      $result=mysql_query($query) or die("error...");
      echo "Update Successful\n";
      echo "<script>exit();</script>";

    }
?>
</form>
</html>
```

```
<html>
   <head>
      <? include("include/include.php"); ?>
      <title>
         <? //title is "Search" if search field is null, "Results" otherwise
            if ($search == "") {
               echo("Search");
            } else if (!($search == "")) {
               echo("Results");
            }
         ?>
      </title>
      <? // Performing SQL query
         if ($search == "") {
            $query = "SELECT * FROM tblemp WHERE empActCode=\"0\"";
         } else {
            $query = "SELECT * FROM tblemp WHERE empLastName=\"".$search."\" AND empActCode=\"0\"";
         }
         $result = mysql_query($query) or die("Query failed");
      ?>
   </head>

   <body bgcolor="#FFFFFF" text="#000000" onLoad="search.search.focus();search.search.select();">
   <b>Current Employee Listings</b>
   <form name="search" method="get" action="">
     <div align="right"> <b>Search</b>
       <input type="text" name="search" value=<?echo"\"$search\"";?>>
       <INPUT TYPE="submit" value="Search">
     </div>
   </form>
   <hr size="1" width="95%">
   <table width="97%" border="0">
     <tr>
       <td><B>Name</B></td>
       <td><B>Department</B></td>
       <td><B>Job Title</B></td>
       <td><B>Office Phone</B></td>
       <td><B>Mail Code</B></td>
       <td><B>Room</B></td>
     </tr>
      <? // print results
         while ($line = mysql_fetch_assoc($result)) {
            print " <tr>\n";
            print "  <td><a href='employeesubmit.php?empID=".$line["empID"]."'>$line[empLastName], $line[empFirstName]</a></td>\n";
            print "  <td>" . getData("deptDesc","deptID","tblDept",$line["empDeptID"],"Department Name") . "</td>\n";
            print "  <td>" . getData("jobDesc","jobID","tblJobDesc",$line["empJobID"],"Job Title") . "</td>\n";
            print "  <td>" . parsePhone($line["empWorkPhone"]) . "</td>\n";
            print "  <td>";
               $tmpquery = "SELECT * FROM tblMailStop WHERE msID = ".$line["empMailStopID"];
               $tmpresult = mysql_query($tmpquery) or die("Query failed when getting mailstop");
               while ($tmpline = mysql_fetch_assoc($tmpresult))
                  echo $tmpline[msDesc];
            print "</td>\n";
            print "  <td>". $line["empRoom"]."</td>\n";
            print " </tr>\n";
         }
      ?>
   </table>
```

```
    <? // Closing connection
       mysql_close($link);
    ?>
    </body>
</html>
```

```
<html>
<head>
   <? include("include/include.php"); ?>
<body>
   <? // update employee
      $query="UPDATE tblEmp SET ";
         $query=$query."`empActCode`= $actCode";
      $query=$query." WHERE `empID` = $id";
      $result=mysql_query($query) or die("error...");
   ?>
   <div align="center">
   <br><font size=+1><?echo ucfirst($empFirstName)." ".ucfirst($empLastName)?>'s status updated to
   <?if ($actCode=="0")
      echo "active.";
   else
      echo "inactive.";
   ?>
   <br><br></font><font size=-1><i>
   Return to:
       <a href=# onclick="history.back();">Activity Listings</a>
       <a href=# onclick="window.location='main.htm';">Main Page</a>
   </font></i>
   </div>

</body>
</html>
```